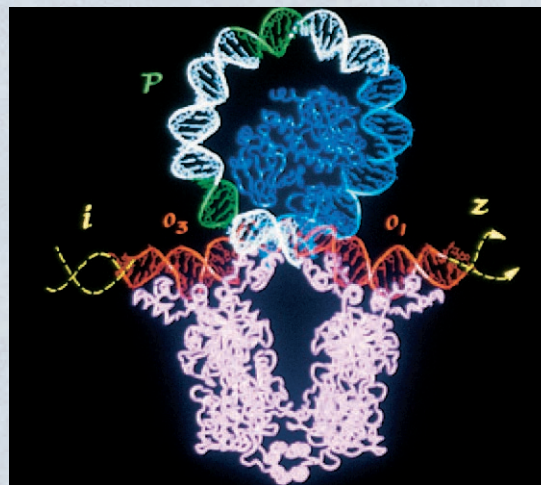


CHAPTER 12

Genes: Expression and Regulation



Lactose operon activity is under the control of a repressor protein. The *lac* repressor (violet) and catabolite activator protein (blue) are bound to the *lac* operon. The repressor blocks transcription when bound to the operators (red).

Outline

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Concepts

1. In transcription the RNA polymerase copies the appropriate sequence on the DNA template strand to produce a complementary RNA copy of the gene. Transcription differs in a number of ways between prokaryotes and eukaryotes, even though the basic mechanism of RNA polymerase action is essentially the same.
2. Translation is the process by which the nucleotide sequence of mRNA is converted into the amino acid sequence of a polypeptide through the action of ribosomes, tRNAs, aminoacyl-tRNA synthetases, ATP and GTP energy, and a variety of protein factors. As in the case of DNA replication, this complex process is designed to minimize errors.
3. The long-term regulation of metabolism in bacteria is achieved through the control of transcription by such mechanisms as sigma factors, repressor proteins during induction and repression, and by the attenuation of many biosynthetic operons.
4. Prokaryotes must be able to respond rapidly to changing environmental conditions and often control many operons simultaneously using global regulatory systems.
5. DNA replication and cell division are coordinated in such a way that the distribution of new DNA copies to each daughter cell is ensured.

The particular field which excites my interest is the division between the living and the non-living, as typified by, say, proteins, viruses, bacteria and the structure of chromosomes. The eventual goal, which is somewhat remote, is the description of these activities in terms of their structure, i.e., the spatial distribution of their constituent atoms, in so far as this may prove possible. This might be called the chemical physics of biology.

—Francis Crick

Chapter 11 is concerned with the genetic material of microorganisms. Its focus is on the structure and replication of DNA, the nature of the genetic code and genes, and the way in which genes change by mutation. This provides background for understanding chapters 12 through 15.

Chapter 12 is devoted to genetic expression and its regulation. An adequate knowledge of transcription and protein synthesis is essential to understanding molecular biology and microbial genetics. Thus the first two sections describe DNA transcription and protein synthesis in some detail. Because the role of these processes in the life of microorganisms depends on their proper regulation, the second half of the chapter is devoted to regulation. The control of mRNA synthesis and attenuation are discussed first. Then more complex levels of regulation are described in sections on global regulatory systems, two-component phosphorelay systems, and control of the bacterial cell cycle.

12.1 DNA Transcription or RNA Synthesis

As mentioned earlier, synthesis of RNA under the direction of DNA is called transcription. The RNA product has a sequence complementary to the DNA template directing its synthesis (**table 12.1**). Thymine is not normally found in mRNA and rRNA. Although ade-

Table 12.1 RNA Bases Coded for by DNA

DNA Base	Purine or Pyrimidine Incorporated into RNA
Adenine	Uracil
Guanine	Cytosine
Cytosine	Guanine
Thymine	Adenine

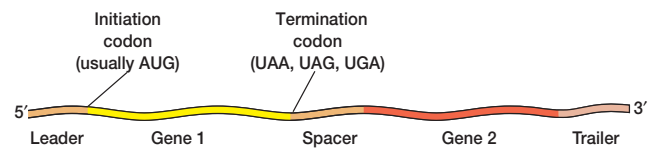


Figure 12.1 A Polygenic Bacterial Messenger RNA. See text for details.

nine directs the incorporation of thymine during DNA replication, it usually codes for uracil during RNA synthesis. Transcription generates three kinds of RNA. Messenger RNA (mRNA) bears the message for protein synthesis. **Transfer RNA (tRNA)** carries amino acids during protein synthesis, and **ribosomal RNA (rRNA)** molecules are components of ribosomes. The structure and synthesis of prokaryotic mRNA is described first.

Transcription in Prokaryotes

Prokaryotic mRNA is a single-stranded RNA of variable length containing directions for the synthesis of one to many polypeptides. Messenger RNA molecules also have sequences that do not code for polypeptides (**figure 12.1**). There is a nontranslated **leader sequence** of 25 to 150 bases at the 5' end preceding the initiation codon. In addition, polygenic mRNAs (those directing the synthesis of more than one polypeptide) have spacer regions separating the segments coding for individual polypeptides. Polygenic messenger polypeptides usually function together in some way (e.g., as part of the same metabolic pathway). At the 3' end, following the last termination codon, is a nontranslated trailer.

Messenger RNA is synthesized under the direction of DNA by the enzyme **RNA polymerase**. An *E. coli* cell can have as many as 7,000 RNA polymerase molecules; only 2,000 to 5,000 polymerases may be active at any one time. The reaction is quite similar to that catalyzed by DNA polymerase. ATP, GTP, CTP, and UTP are used to produce an RNA copy of the DNA sequence. As mentioned earlier, these nucleotides contain ribose rather than deoxyribose.



RNA synthesis, like DNA synthesis, proceeds in a 5' to 3' direction with new nucleotides being added to the 3' end of the growing chain at a rate of about 40 nucleotides per second at 37°C (**figure 12.2**). The RNA polymerase opens or unwinds the double helix to form a

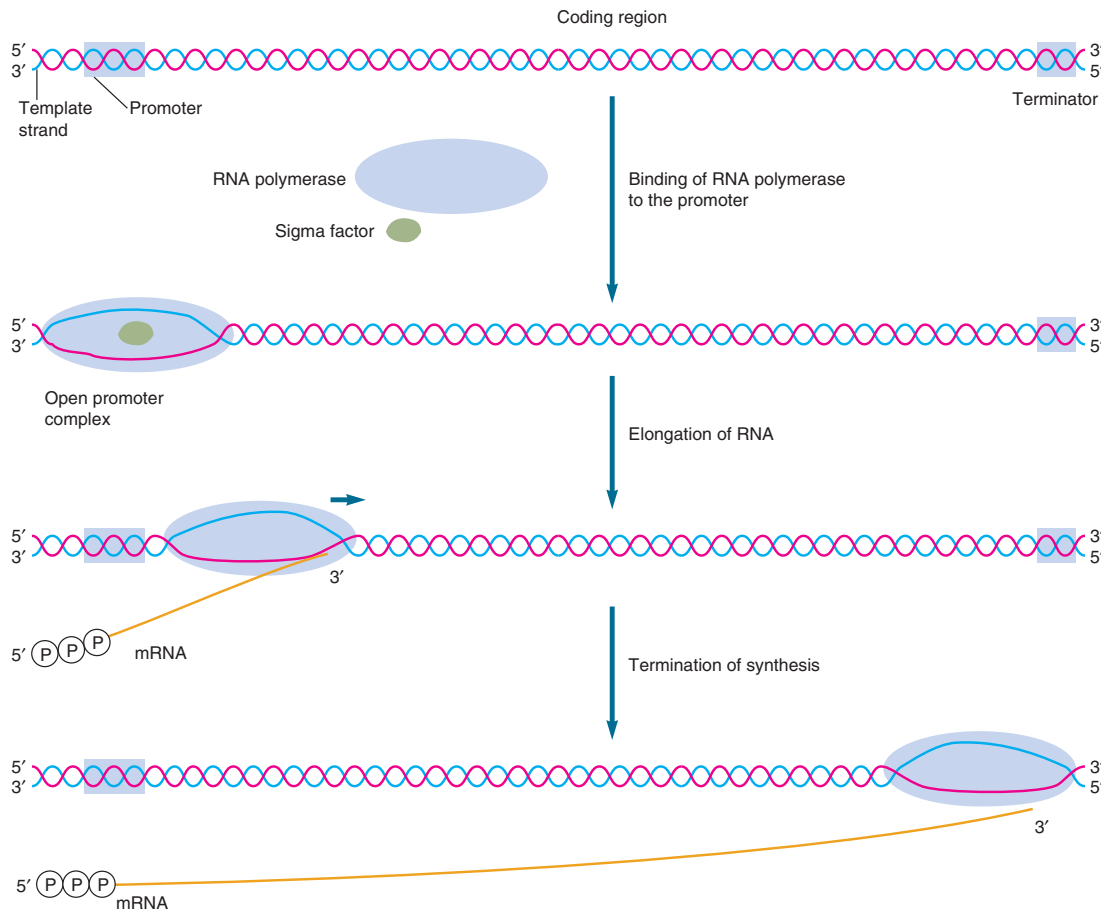


Figure 12.2 mRNA Transcription From DNA. The lower DNA strand directs mRNA synthesis, and the RNA polymerase moves from left to right. Transcription has been simplified for clarity and divided into three general phases: binding of RNA polymerase to the promoter with the aid of a sigma factor, synthesis of RNA by the polymerase under the direction of the template strand, and termination of the process coupled with release of the RNA product. Upon binding, the RNA polymerase unwinds DNA to form a transcription bubble or open complex so that it can copy the template strand. See text for details.

transcription bubble, about 12 to 20 base pairs in length, and transcribes the template strand to produce an RNA transcript that is complementary and antiparallel to the DNA template. It should be noted that pyrophosphate is produced in both DNA and RNA polymerase reactions. Pyrophosphate is then removed by hydrolysis to orthophosphate in a reaction catalyzed by the pyrophosphatase enzyme. Removal of the pyrophosphate product makes DNA and RNA synthesis irreversible. If the pyrophosphate level were too high, DNA and RNA would be degraded by a reversal of the polymerase reactions.

The RNA polymerase of *E. coli* is a very large molecule (about 480,000 daltons) containing four types of polypeptide chains: α , β , β' , and σ . The **core enzyme** is composed of four chains (α_2 , β , β') and catalyzes RNA synthesis. The **sigma factor** (σ) has no catalytic activity but helps the core enzyme recognize the start of genes. Once RNA synthesis begins, the sigma factor dissociates from the core enzyme–DNA complex and is available to aid another core enzyme. There are several different

sigma factors in *E. coli*; σ^{70} (about 70,000 molecular weight) is most often involved in transcription initiation. The precise functions of the α , β , and β' polypeptides are not yet clear. The α subunit seems to be involved in the assembly of the core enzyme, recognition of promoters (see below), and interaction with some regulatory factors. The binding site for DNA is on β' , and the β subunit binds ribonucleotide substrates. Rifampin, an RNA polymerase inhibitor, binds to the β subunit.

The region to which RNA polymerase binds with the aid of the sigma factor is called the **promoter**. The procaryotic promoter sequence is not transcribed. A 6 base sequence (usually TTGACA), approximately 35 base pairs before the transcription starting point, is present in *E. coli* promoters. A TATAAT sequence or **Pribnow box** lies within the promoter about 10 base pairs before the starting point of transcription or around 16 to 18 base pairs from the first hexamer sequence. The RNA polymerase recognizes these sequences, binds to the promoter, and unwinds a short segment of

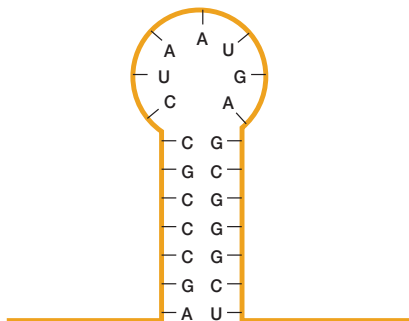


Figure 12.3 Prokaryotic Terminators. An example of a hairpin structure formed by an mRNA terminator sequence.

DNA beginning around the Pribnow box. Transcription starts 6 or 7 base pairs away from the 3' end of the promoter. The RNA polymerase remains at the promoter while it constructs a chain about 9 nucleotides long, then it begins to move down the template strand. The first base used in RNA synthesis is usually a purine, either ATP or GTP. Since these phosphates are not removed during transcription, the 5' end of procaryotic mRNA has a triphosphate attached to the ribose. [Promoter structure and function \(pp. 242–44\)](#)

There also must be stop signals to mark the end of a gene or sequence of genes and stop transcription by the RNA polymerase. Procaryotic **terminators** often contain a sequence coding for an RNA stretch that can hydrogen bond to form a hairpin-shaped loop and stem structure (**figure 12.3**). This structure appears to cause the RNA polymerase to pause or stop transcribing DNA. There are two kinds of stop signals or terminators. The first type contains a stretch of about six uridine residues following the mRNA hairpin and causes the polymerase to stop transcription and release the mRNA without the aid of any accessory factors. The second kind of terminator lacks a poly-U region, and often the hairpin; it requires the aid of a special protein, the **rho factor** (ρ). It is thought that rho binds to mRNA and moves along the molecule until it reaches the RNA polymerase that has halted at a terminator. The rho factor then causes the polymerase to dissociate from the mRNA, probably by unwinding the mRNA-DNA complex.

Transcription in Eucaryotes

Transcriptional processes in eucaryotic microorganisms (and in other eucaryotic cells) differ in several ways from procaryotic transcription. There are three major RNA polymerases, not one as in procaryotes. RNA polymerase II, associated with chromatin in the nuclear matrix, is responsible for mRNA synthesis. Polymerases I and III synthesize rRNA and tRNA, respectively (**table 12.2**). The eucaryotic RNA polymerase II is a large aggregate, at least 500,000 daltons in size, with about 10 or more subunits. It is inhibited by the octapeptide α -amanitin. Unlike the bacterial polymerase it requires extra transcription factors to recognize its promoters. The polymerase binds near the start point; the transcription factors bind to the rest of the promoter. Eucaryotic promoters also differ from those in procaryotes. They have combinations of several elements. Three of

Table 12.2 Eucaryotic RNA Polymerases

Enzyme	Location	Product
RNA polymerase I	Nucleolus	rRNA (5.8S, 18S, 28S)
RNA polymerase II	Chromatin, nuclear matrix	mRNA
RNA polymerase III	Chromatin, nuclear matrix	tRNA, 5S rRNA

the most common are the TATA box (located about 30 base pairs before the start point or upstream), the CAAT box (about 75 base pairs upstream), and the GC box (90 base pairs upstream). Recently it has been shown that the TATA-binding protein sharply bends the DNA on attachment. This makes the DNA more accessible to other initiation factors. A variety of general transcription factors, promoter specific factors, and promoter elements have been discovered in different eucaryotic cells. Each eucaryotic gene seems to be regulated differently, and more research will be required to understand the regulation of eucaryotic gene transcription.

Eucaryotic mRNA arises from **posttranscriptional modification** of large RNA precursors, about 5,000 to 50,000 nucleotides long, called **heterogeneous nuclear RNA (hnRNA)** molecules. These are the products of RNA polymerase II activity (**figure 12.4**). After hnRNA synthesis, the precursor RNA is cleaved by an endonuclease to yield the proper 3'-OH group. The enzyme polyadenylate polymerase then catalyzes the addition of adenylic acid to the 3' end of hnRNA to produce a poly-A sequence about 200 nucleotides long. The hnRNA finally is cleaved to generate the functional mRNA. Usually eucaryotic mRNA also differs in having a 5' cap consisting of 7-methylguanosine attached to the 5'-hydroxyl by a triphosphate linkage (**figure 12.5**). The adjacent nucleotide also may be methylated.

Eucaryotic mRNAs have 5' caps, unlike procaryotic mRNAs. Both types of cells can have mRNA with 3' poly-A, but procaryotes have poly-A much less often and the tracts are shorter. In addition, eucaryotic mRNA normally is monogenic in contrast to procaryotic mRNA, which often contains transcripts of two or more genes. The functions of poly-A and capping still are not completely clear. Poly-A protects mRNA from rapid enzymatic degradation. The poly-A tail must be shortened to about 10 nucleotides before mRNA can be degraded. Poly-A also seems to aid in mRNA translation. The 5' cap on eucaryotic messengers may promote the initial binding of ribosomes to the messenger. The cap also may protect the messenger from enzymatic attack.

Many eucaryotic genes differ from procaryotic genes in being split or interrupted, which leads to another type of posttranscriptional processing. **Split or interrupted genes** have **exons** (expressed sequences), regions coding for RNA that end up in the final RNA product (e.g., mRNA). Exons are separated from one another by **introns** (intervening sequences), sequences coding for RNA that is missing from the final product (**figure 12.4b**). The initial RNA transcript has the intron sequences present in the interrupted gene. Genes coding for rRNA and tRNA may also be interrupted. Except for cyanobacteria and Archaea (*see chapters 20 and 21*), interrupted genes have not been found in procaryotes.

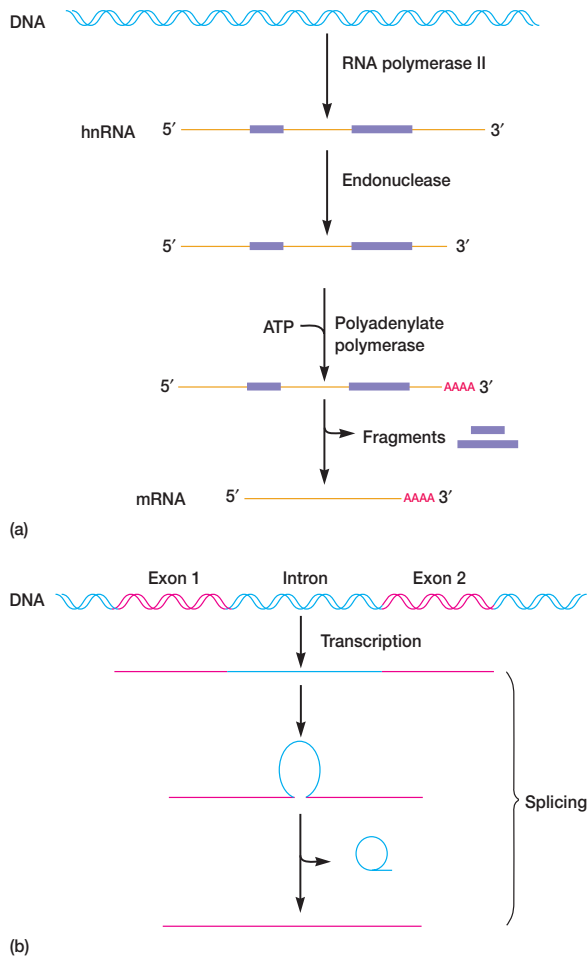


Figure 12.4 Eucaryotic mRNA Synthesis. (a) The production of eucaryotic messenger RNA. The addition of poly-A to the 3' end of mRNA is included, but not the capping of the 5' end. Poly-A sequence and introns are in color. (b) The splicing of interrupted genes to produce mRNA. Poly-A sequences and exons are in color. The excised intron is in the shape of a circle or lariat.

Introns are removed from the initial RNA transcript by a process called **RNA splicing** (figure 12.4b). The intron's borders must be clearly marked for accurate removal, and this is the case. Exon-intron junctions have a GU sequence at the intron's 5' boundary and an AG sequence at its 3' end. These two sequences define the splice junctions and are recognized by special RNA molecules. The nucleus contains several **small nuclear RNA (snRNA)** molecules, about 60 to 300 nucleotides long. These complex with proteins to form small *nuclear ribonucleoprotein* particles called snRNPs or snurps. Some of the snRNPs recognize splice junctions and ensure splicing accuracy. For example, U1-snRNP recognizes the 5' splice junction, and U5-snRNP recognizes the 3' junction. Splicing of pre-mRNA occurs in a large complex called a **spliceosome** that contains the pre-mRNA, at least five kinds of snRNPs, and non-snRNP splicing factors.

As just mentioned, a few rRNA genes also have introns. Some of these pre-rRNA molecules are self-splicing. The RNA actually catalyzes the splicing reaction and now is called a **ribozyme** (Box 12.1). Thomas Cech first discovered that pre-rRNA from the ciliate protozoan *Tetrahymena thermophila* is self-splicing. Sidney Altman then showed that ribonuclease P, which cleaves a fragment from one end of pre-tRNA, contains a piece of RNA that catalyzes the reaction. Several other self-splicing rRNA introns have since been discovered. Cech and Altman received the 1989 Nobel Prize in chemistry for these discoveries.

Although the focus has been on mRNA synthesis, it should be noted that both rRNAs and tRNAs also begin as parts of large precursors synthesized by RNA polymerases (table 12.2). The final rRNA and tRNA products result from posttranscriptional processing, as mentioned previously.

1. Define the following terms: leader, trailer, spacer region, polygenic mRNA, RNA polymerase core enzyme, sigma factor, promoter, template strand, terminator, and rho factor.
2. Define or describe posttranscriptional modification, heterogeneous nuclear RNA, 3' poly-A sequence, 5' capping, split or interrupted genes, exon, intron, RNA splicing, snRNA, spliceosome, and ribozyme.

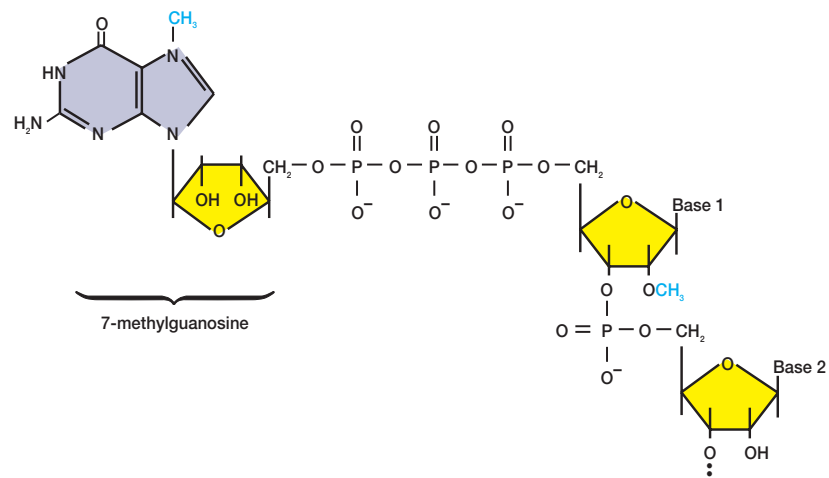


Figure 12.5 The 5' Cap of Eucaryotic mRNA.

Box 12.1

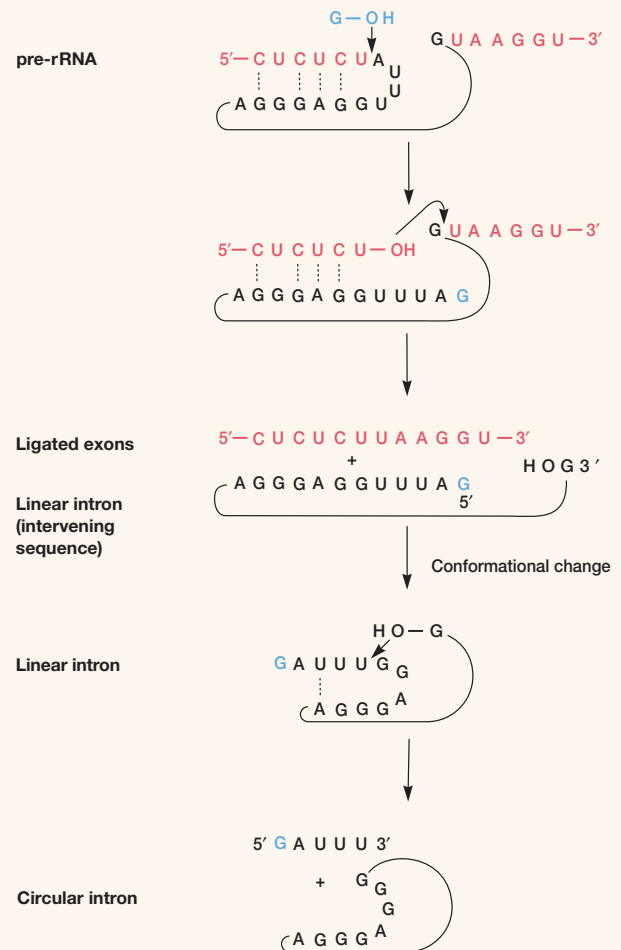
Catalytic RNA (Ribozymes)

Until recently biologists believed that all cellular reactions were catalyzed by proteins called enzymes (*see section 8.6*). The discovery during 1981–1984 by Thomas Cech and Sidney Altman that RNA also can sometimes catalyze reactions has transformed our way of thinking about topics as diverse as catalysis and the origin of life. It is now clear that some RNA molecules, called ribozymes, catalyze reactions that alter either their own structure or that of other RNAs.

This discovery has stimulated scientists to hypothesize that the early earth was an “RNA world” in which RNA acted as both the genetic material and a reaction catalyst. Experiments showing that introns from *Tetrahymena thermophila* can catalyze the formation of polycytidylic acid under certain circumstances have further encouraged such speculations. Some have suggested that RNA viruses are “living fossils” of the original RNA world.

The best-studied ribozyme activity is the self-splicing of RNA. This process is widespread and occurs in *Tetrahymena* pre-rRNA; the mitochondrial rRNA and mRNA of yeast and other fungi; chloroplast tRNA, rRNA, and mRNA; and in mRNA from some bacteriophages (e.g., the T4 phage of *E. coli*). The 413-nucleotide rRNA intron of *T. thermophila* provides a good example of the self-splicing reaction. The reaction occurs in three steps and requires the presence of guanosine (*see Box figure*). First, the 3′-OH group of guanosine attacks the intron’s 5′-phosphate group and cleaves the phosphodiester bond. Second, the new 3′-hydroxyl on the left exon attacks the 5′-phosphate of the right exon. This joins the two exons and releases the intron. Finally, the intron’s 3′-hydroxyl attacks the phosphate bond of the nucleotide 15 residues from its end. This releases a terminal fragment and cyclizes the intron. Self-splicing of this rRNA occurs about 10 billion times faster than spontaneous RNA hydrolysis. Just as with enzyme proteins, the RNA’s shape is essential to catalytic efficiency. The ribozyme even has Michaelis-Menten kinetics (pp. 162–63).

The discovery of ribozymes has many potentially important practical consequences. Ribozymes act as “molecular scissors” and will enable researchers to manipulate RNA easily in laboratory experiments. It also might be possible to protect hosts by specifically removing RNA from pathogenic viruses, bacteria, and fungi. For example, ribozymes are already being tested against the AIDS, herpes, and tobacco mosaic viruses.



Ribozyme Action. The mechanism of *Tetrahymena thermophila* pre-rRNA self-splicing. See text for details.

12.2 Protein Synthesis

The final step in gene expression is protein synthesis or translation. The mRNA nucleotide sequence is translated into the amino acid sequence of a polypeptide chain in this step. Polypeptides are synthesized by the addition of amino acids to the end of the chain with the free α -carboxyl group (the C-terminal end). That is, the synthesis of polypeptides begins with the amino acid at the end of the chain with a free amino group (the N-terminal) and moves in the C-terminal direction. The ribosome is the site of protein synthesis. Protein synthesis is not

only quite accurate but also very rapid. In *E. coli* synthesis occurs at a rate of at least 900 residues per minute; eucaryotic translation is slower, about 100 residues per minute. [Polypeptide and protein structure \(appendix I\)](#)

Many bacteria grow so quickly that each mRNA must be used with great efficiency to synthesize proteins at a sufficiently rapid rate. Ribosomal subunits are free in the cytoplasm if protein is not being synthesized. They come together to form the complete ribosome only when translation occurs. Frequently bacterial mRNAs are simultaneously complexed with several ribosomes, each ribosome reading the mRNA message and synthesizing a polypeptide. At

maximal rates of mRNA use, there may be a ribosome every 80 nucleotides along the messenger or as many as 20 ribosomes simultaneously reading an mRNA that codes for a 50,000 dalton polypeptide. A complex of mRNA with several ribosomes is called a **polyribosome** or polysome. Polysomes are present in both procaryotes and eucaryotes. Bacteria can further increase the efficiency of gene expression through coupled transcription and translation (**figure 12.6**). While RNA polymerase is synthesizing an

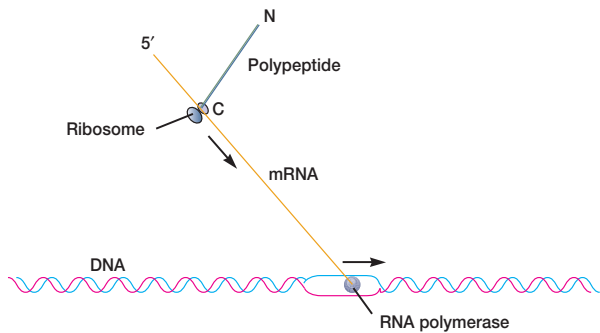


Figure 12.6 Coupled Transcription and Translation in Bacteria. mRNA is synthesized 5' to 3' by RNA polymerase while ribosomes are attaching to the newly formed 5' end of mRNA and translating the message even before it is completed. Polypeptides are synthesized in the N-terminal to C-terminal direction.

mRNA, ribosomes can already be attached to the messenger and involved in polypeptide synthesis. Coupled transcription and translation is possible in procaryotes because a nuclear envelope does not separate the translation machinery from DNA as it does in eucaryotes (*see figure 3.14*).

Transfer RNA and Amino Acid Activation

The first stage of protein synthesis is **amino acid activation**, a process in which amino acids are attached to transfer RNA molecules. These RNA molecules are normally between 73 and 93 nucleotides in length and possess several characteristic structural features. The structure of tRNA becomes clearer when its chain is folded in such a way to maximize the number of normal base pairs, which results in a cloverleaf conformation of five arms or loops (**figure 12.7**). The acceptor or amino acid stem holds the activated amino acid on the 3' end of the tRNA. The 3' end of all tRNAs has the same —C—C—A sequence; the amino acid is attached to the terminal adenylic acid. At the other end of the cloverleaf is the anticodon arm, which contains the **anticodon triplet** complementary to the mRNA codon triplet. There are two other large arms: the D or DHU arm has the unusual pyrimidine nucleoside dihydrouridine; and the T or TΨC arm has ribothymidine (T) and pseudouridine (Ψ), both of which are unique to tRNA. Finally, the cloverleaf has a variable arm whose length changes with the overall length of the tRNA; the other arms are fairly constant in size.

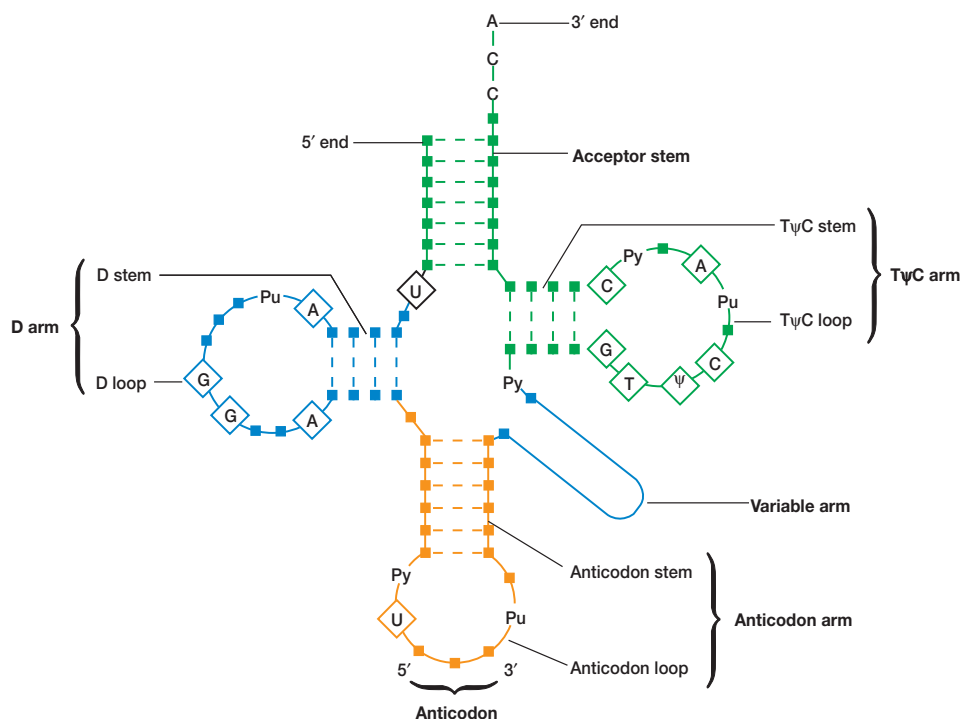


Figure 12.7 tRNA Structure. The cloverleaf structure for tRNA in procaryotes and eucaryotes. Bases found in all tRNAs are in diamonds; purine and pyrimidine positions in all tRNAs are labeled Pu and Py respectively.

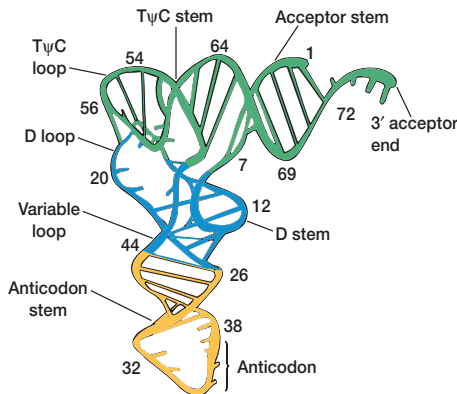
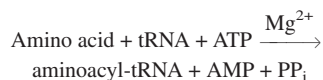


Figure 12.8 Transfer RNA Conformation. The three-dimensional structure of tRNA. The various regions are distinguished with different colors.

Transfer RNA molecules are folded into an L-shaped structure (**figure 12.8**). The amino acid is held on one end of the L, the anticodon is positioned on the opposite end, and the corner of the L is formed by the D and T loops. Because there must be at least one tRNA for each of the 20 amino acids incorporated into proteins, at least 20 different tRNA molecules are needed. Actually more tRNA species exist (*see pp. 240–41*).

Amino acids are activated for protein synthesis through a reaction catalyzed by **aminoacyl-tRNA synthetases** (**figure 12.9**).



Just as is true of DNA and RNA synthesis, the reaction is driven to completion when the pyrophosphate product is hydrolyzed to two orthophosphates. The amino acid is attached to the 3'-hydroxyl of the terminal adenylic acid on the tRNA by a high-energy bond (**figure 12.10**), and is readily transferred to the end of a growing peptide chain. This is why the amino acid is called activated.

There are at least 20 aminoacyl-tRNA synthetases, each specific for a single amino acid and for all the tRNAs (cognate tRNAs) to which each may be properly attached. This specificity is critical because once an incorrect acid is attached to a tRNA, it will be incorporated into a polypeptide in place of the correct amino acid. The protein synthetic machinery recognizes only the anticodon of the aminoacyl-tRNA and cannot tell whether the correct amino acid is attached. Some aminoacyl-tRNA synthetases will even proofread just like DNA polymerases do. If the wrong aminoacyl-tRNA is formed, aminoacyl-tRNA synthetases will hydrolyze the amino acid from the tRNA rather than release the incorrect product.

The Ribosome

The actual process of protein synthesis takes place on ribosomes that serve as workbenches, with mRNA acting as the blueprint. Prokaryotic ribosomes have a sedimentation value of 70S and a mass of 2.8 million daltons. A rapidly growing *E. coli* cell may

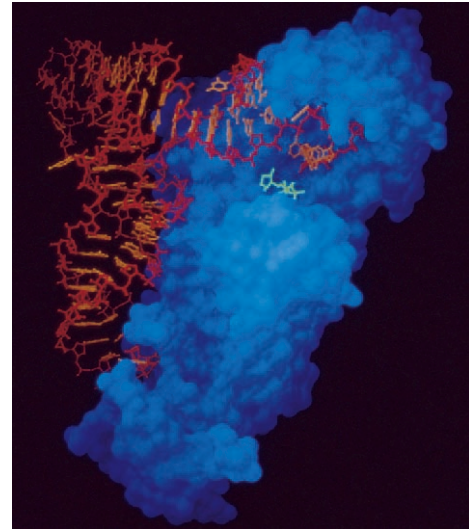


Figure 12.9 An Aminoacyl-tRNA Synthetase. A model of *E. coli* glutamyl-tRNA synthetase complexed with its tRNA and ATP. The enzyme is in blue, the tRNA in red and yellow, and ATP in green.

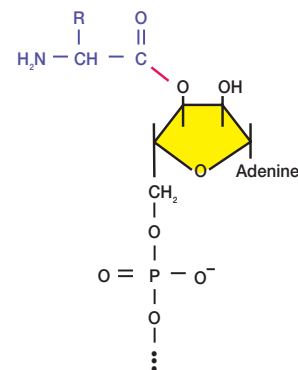


Figure 12.10 Aminoacyl-tRNA. The 3' end of an aminoacyl-tRNA. The activated amino acid is attached to the 3'-hydroxyl of adenylic acid by a high-energy bond (red).

have as many as 15,000 to 20,000 ribosomes, about 15% of the cell mass. [Introduction to ribosomal function and the Svedberg unit \(p. 52\)](#)

The prokaryotic ribosome is an extraordinarily complex organelle made of a 30S and a 50S subunit (**figure 12.11**). Each subunit is constructed from one or two rRNA molecules and many polypeptides. The shape of ribosomal subunits and their association to form the 70S ribosome are depicted in **figure 12.12**. The region of the ribosome directly responsible for translation is called the translational domain (**figure 12.12d**). Both subunits contribute to this domain, located in the upper half of the small subunit and in the associated areas of the large subunit. For example, the peptidyl transferase (p. 270) is found on the central protuberance of the large subunit. The growing peptide chain

emerges from the large subunit at the exit domain. This is located on the side of the subunit opposite the central protuberance in both procaryotes and eucaryotes.

Eucaryotic cytoplasmic ribosomes are 80S, with a mass of 4 million daltons, and are composed of two subunits, 40S and 60S. Many of these ribosomes are found free in the cytoplasmic matrix, whereas others are attached to membranes of the endoplasmic reticulum by their 60S subunit at a site next to the exit domain. The ribosomes of eucaryotic mitochondria and chloroplasts are smaller than cytoplasmic ribosomes and resemble the procaryotic organelle.

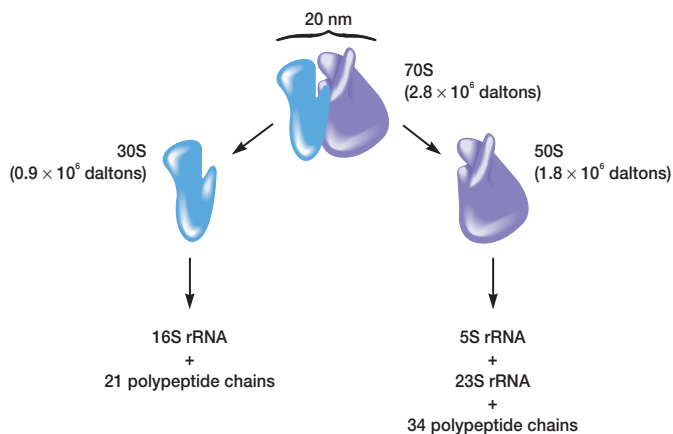


Figure 12.11 The 70S Ribosome. The structure of the procaryotic ribosome.

Ribosomal RNA is thought to have two roles. It obviously contributes to ribosome structure. The 16S rRNA of the 30S subunit also may aid in the initiation of protein synthesis in procaryotes. There is evidence that the 3' end of the 16S rRNA complexes with an initiating signal site on the mRNA and helps position the mRNA on the ribosome. It also binds initiation factor 3 (p. 270) and the 3' CCA end of aminoacyl-tRNA. Because of the discovery of catalytic RNA, some have proposed that ribosomal RNA has a catalytic role in protein synthesis. [The use of 16S rRNA sequences in the study of phylogeny \(pp. 433–35\)](#)

Initiation of Protein Synthesis

Protein synthesis proper may be divided into three stages: initiation, elongation, and termination.

In the initiation stage *E. coli* and most bacteria begin protein synthesis with a specially modified aminoacyl-tRNA, *N*-formylmethionyl-tRNA^{fMet} (figure 12.13). Because the α -amino is blocked by a formyl group, this aminoacyl-tRNA can be used only for initiation. When methionine is to be added to a growing polypeptide chain, a normal methionyl-tRNA^{Met} is employed. Eucaryotic protein synthesis (except in the mitochondrion and chloroplast) and archaeal protein synthesis begin with a special initiator methionyl-tRNA^{Met}. Although most bacteria start protein synthesis with formylmethionine, the formyl group does not remain but is hydrolytically removed. In fact, one to three amino acids may be removed from the amino terminal end of the polypeptide after synthesis.

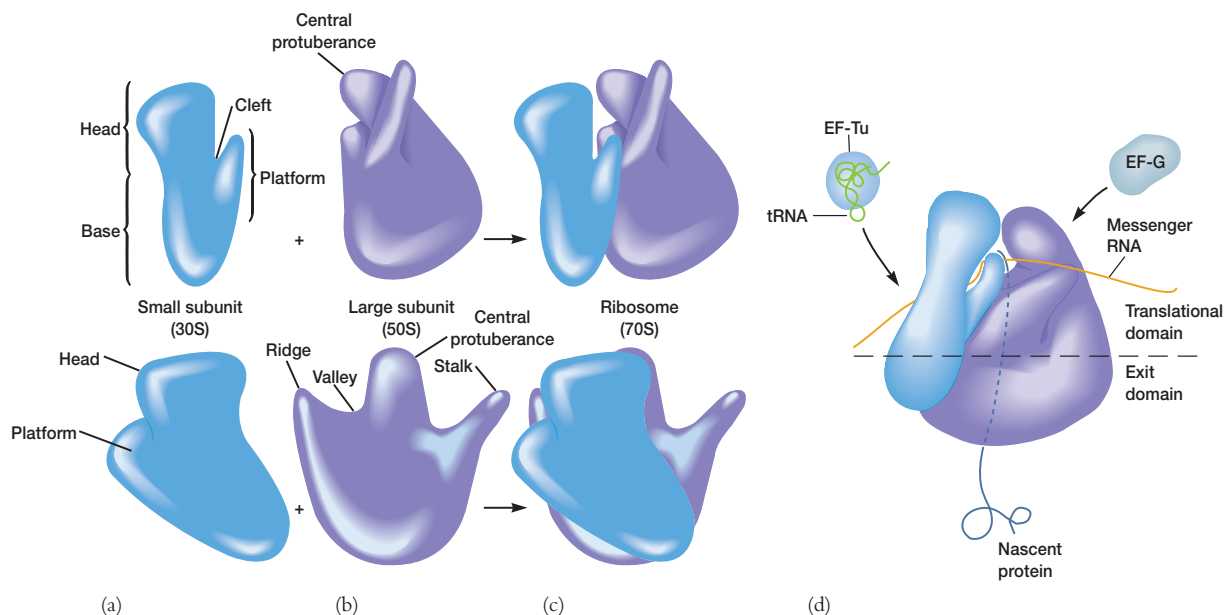


Figure 12.12 Two Views of the *E. coli* Ribosome. (a) The 30S subunit. (b) The 50S subunit. (c) The complete 70S ribosome. (d) Diagrammatic representation of ribosomal structure with the translational and exit domains shown. The locations of elongation factor and mRNA binding are shown. The growing peptide chain probably remains unfolded and extended until it leaves the large subunit.

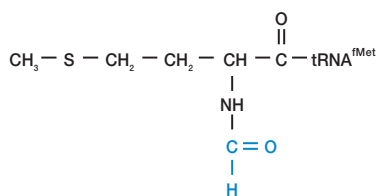


Figure 12.13 Procaryotic Initiator tRNA. The initiator aminoacyl-tRNA, *N*-formylmethionyl-tRNA^{fMet}, is used by bacteria. The formyl group is in color. Archaea use methionyl-tRNA for initiation.

Figure 12.14 shows the initiation process in procaryotes. The initiator *N*-formylmethionyl-tRNA^{fMet} (fMet-tRNA) binds to the free 30S subunit first. Next mRNA attaches to the 30S subunit and is positioned properly through interactions with both the 3' end of the 16S rRNA and the anticodon of fMet-tRNA. Messengers have a special **initiator codon** (AUG or sometimes GUG) that specifically binds with the fMet-tRNA anticodon (see section 12.2). Finally, the 50S subunit binds to the 30S subunit-mRNA forming an active ribosome-mRNA complex. The fMet-tRNA is positioned at the peptidyl or P site (see description of the elongation cycle).

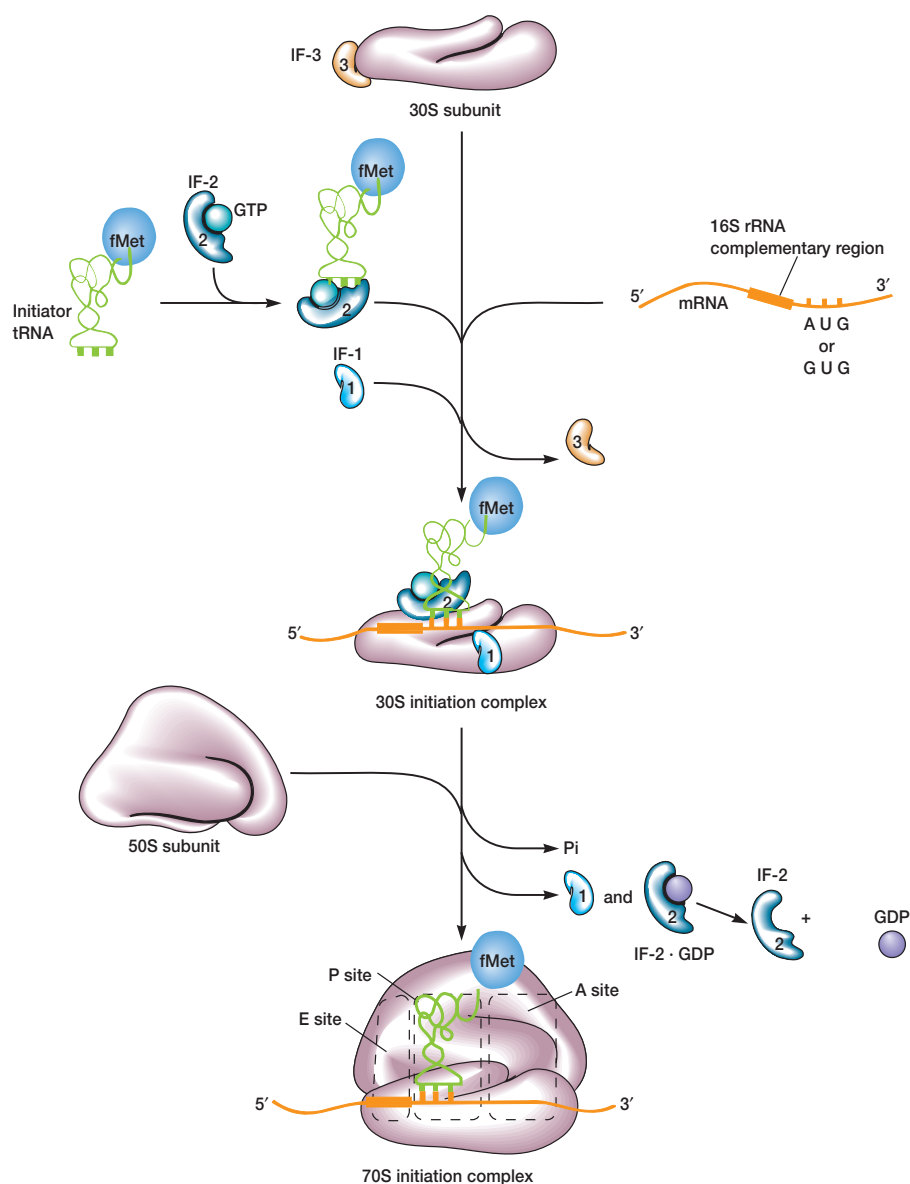


Figure 12.14 Initiation of Protein Synthesis. The initiation of protein synthesis in procaryotes. The following abbreviations are employed: IF-1, IF-2, and IF-3 stand for initiation factors 1, 2, and 3; initiator tRNA is *N*-formylmethionyl-tRNA^{fMet}. The ribosomal locations of initiation factors are depicted for illustration purposes only. They do not represent the actual initiation factor binding sites. See text for further discussion.

There is some uncertainty about the exact initiation sequence, and mRNA may bind before fMet-tRNA in procaryotes. Eucaryotic initiation appears to begin with the binding of a special initiator Met-tRNA to the small subunit, followed by attachment of the mRNA.

In procaryotes three protein **initiation factors** are required (figure 12.14). Initiation factor 3 (IF-3) prevents 30S subunit binding to the 50S subunit and promotes the proper mRNA binding to the 30S subunit. IF-2, the second initiation factor, binds GTP and fMet-tRNA and directs the attachment of fMet-tRNA to the 30S subunit. GTP is hydrolyzed during association of the 50S and 30S subunits. The third initiation factor, IF-1, appears to be needed for release of IF-2 and GDP from the completed 70S ribosome. IF-1 also may aid in the binding of the 50S subunit to the 30S subunit. Eucaryotes require more initiation factors; otherwise the process is quite similar to that of procaryotes.

The initiation of protein synthesis is very elaborate. Apparently the complexity is necessary to ensure that the ribosome does not start synthesizing a polypeptide chain in the middle of a gene—a disastrous error.

Elongation of the Polypeptide Chain

Every amino acid addition to a growing polypeptide chain is the result of an **elongation cycle** composed of three phases: aminoacyl-tRNA binding, the transpeptidation reaction, and translocation. The process is aided by special protein **elongation factors** (just as with the initiation of protein synthesis). In each turn of the cycle, an amino acid corresponding to the proper mRNA codon is added to the C-terminal end of the polypeptide chain. The procaryotic elongation cycle is described next.

The ribosome has three sites for binding tRNAs: (1) the **peptidyl** or **donor site** (the **P site**), (2) the **aminoacyl** or **acceptor site** (the **A site**), and (3) the **exit site** (the **E site**). At the beginning of an elongation cycle, the peptidyl site is filled with either *N*-formylmethionyl-tRNA^{fMet} or peptidyl-tRNA and the aminoacyl and exit sites are empty (**figure 12.15**). Messenger RNA is bound to the ribosome in such a way that the proper codon interacts with the P site tRNA (e.g., an AUG codon for fMet-tRNA). The next codon (green) is located within the A site and is ready to direct the binding of an aminoacyl-tRNA.

The first phase of the elongation cycle is the aminoacyl-tRNA binding phase. The aminoacyl-tRNA corresponding to the green codon is inserted into the A site. GTP and the elongation factor EF-Tu, which donates the aminoacyl-tRNA to the ribosome, are required for this insertion. When GTP is bound to EF-Tu, the protein is in its active state and delivers aminoacyl-tRNA to the A site. This is followed by GTP hydrolysis, and the EF-Tu·GDP complex leaves the ribosome. EF-Tu·GDP is converted to EF-Tu·GTP with the aid of a second elongation factor, EF-Ts. Subsequently another aminoacyl-tRNA binds to EF-Tu·GTP (figure 12.15).

Aminoacyl-tRNA binding to the A site initiates the second phase of the elongation cycle, the **transpeptidation reaction** (figure 12.15 and **figure 12.16**). This is catalyzed by the **peptidyl transferase**, located on the 50S subunit. The α -amino group of the A site amino acid nucleophilically attacks the α -carboxyl group of the C-terminal amino acid on the P site tRNA in this reaction (figure 12.16). The peptide chain grows by one amino acid and is trans-

ferred to the A site tRNA. No extra energy source is required for peptide bond formation because the bond linking an amino acid to tRNA is high in energy. Recent evidence strongly suggests that 23S rRNA contains the peptidyl transferase function. Almost all protein can be removed from the 50S subunit, leaving the 23S rRNA and protein fragments. The remaining complex still has peptidyl transferase activity. The high-resolution structure of the large subunit has now been obtained by X-ray crystallography. There is no protein in the active site region. A specific adenine base seems to participate in catalyzing peptide bond formation. Thus the 23S rRNA appears to be the major component of the peptidyl transferase and contributes to both A and P site functions.

The final phase in the elongation cycle is **translocation**. Three things happen simultaneously: (1) the peptidyl-tRNA moves from the A site to the P site; (2) the ribosome moves one codon along mRNA so that a new codon is positioned in the A site; and (3) the empty tRNA leaves the P site. Instead of immediately being ejected from the ribosome, the empty tRNA moves from the P site to the E site and then leaves the ribosome. The intricate process requires the participation of the EF-G or translocase protein and GTP hydrolysis. The ribosome changes shape as it moves down the mRNA in the 5' to 3' direction.

Termination of Protein Synthesis

Protein synthesis stops when the ribosome reaches one of three special **nonsense codons**—UAA, UAG, and UGA (**figure 12.17**). Three **release factors** (RF-1, RF-2, and RF-3) aid the ribosome in recognizing these codons. After the ribosome has stopped, peptidyl transferase hydrolyzes the peptide free from its tRNA, and the empty tRNA is released. GTP hydrolysis seems to be required during this sequence, although it may not be needed for termination in procaryotes. Next the ribosome dissociates from its mRNA and separates into 30S and 50S subunits. IF-3 binds to the 30S subunit and prevents it from reassociating with the 50S subunit until the proper stage in initiation is reached. Thus ribosomal subunits associate during protein synthesis and separate afterward. The termination of eucaryotic protein synthesis is similar except that only one release factor appears to be active.

Protein synthesis is a very expensive process. Three GTP molecules probably are used during the elongation cycle, and two ATP high-energy bonds are required for amino acid activation (ATP is converted to AMP rather than to ADP). Therefore five high-energy bonds are required to add one amino acid to a growing polypeptide chain. GTP also is used in initiation and termination of protein synthesis (figures 12.14 and 12.17). Presumably this large energy expenditure is required to ensure the fidelity of protein synthesis. Very few mistakes can be tolerated.

Although the mechanism of protein synthesis is similar in procaryotes and eucaryotes, procaryotic ribosomes differ substantially from those in eucaryotes. This explains the effectiveness of many important chemotherapeutic agents. Either the 30S or the 50S subunit may be affected. For example, streptomycin binding to the 30S ribosomal subunit inhibits protein synthesis and causes mRNA misreading. Erythromycin binds to the 50S subunit and inhibits peptide chain elongation. [The effect of antibiotics on protein synthesis \(pp. 810–11, 817\)](#)

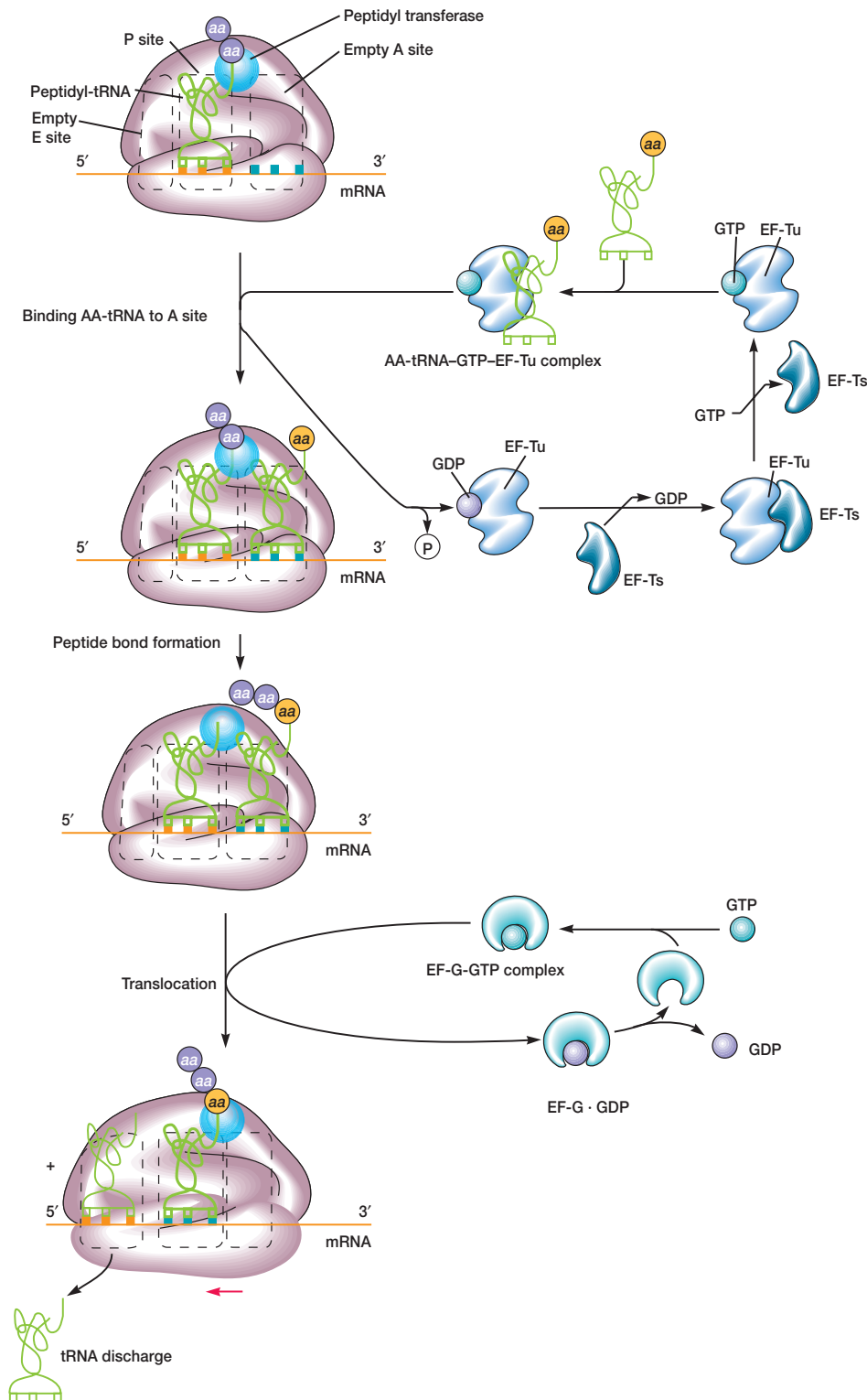


Figure 12.15 Elongation Cycle. The elongation cycle of protein synthesis. The ribosome possesses three sites, a peptidyl or donor site (P site), an aminoacyl or acceptor site (A site), and an exit site (E site). The arrow below the ribosome in translocation step shows the direction of mRNA movement. See text for details.

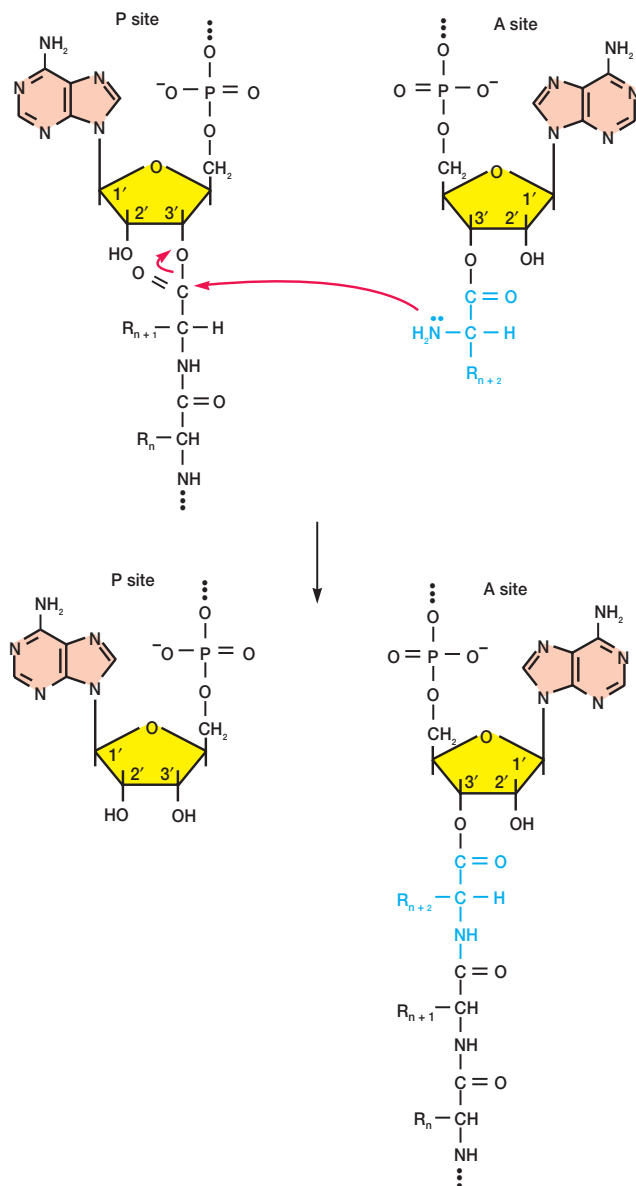


Figure 12.16 Transpeptidation. The peptidyl transferase reaction. The peptide grows by one amino acid and is transferred to the A site.

Protein Folding and Molecular Chaperones

For many years it was believed that polypeptides would spontaneously fold into their final native shape, either as they were synthesized by ribosomes or shortly after completion of protein synthesis. Although the amino acid sequence of a polypeptide does determine its final conformation, it is now clear that special helper proteins aid the newly formed or nascent polypeptide in folding to its proper shape. These proteins, called **molecular chaperones** or chaperones, recognize only unfolded polypeptides or partly denatured proteins and do not bind to normal, func-

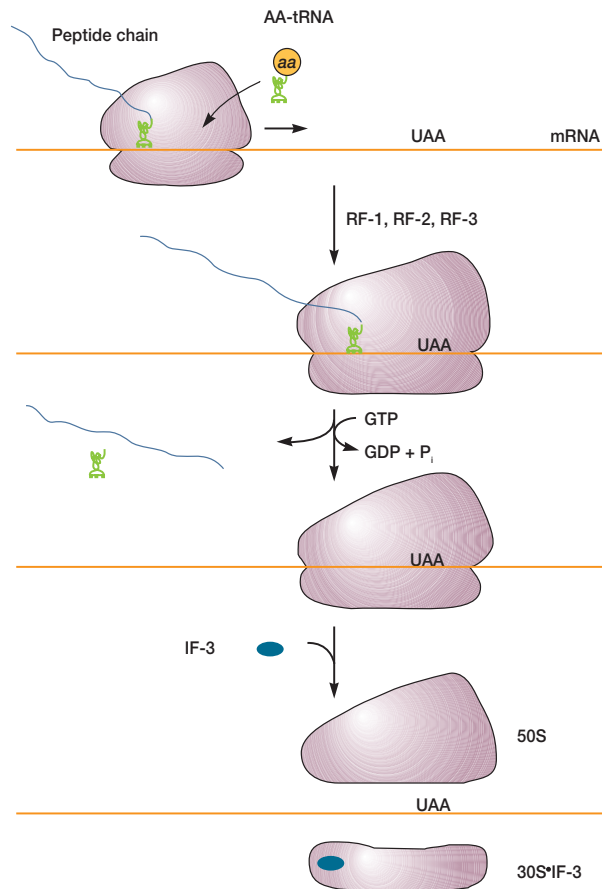


Figure 12.17 Termination of Protein Synthesis in Prokaryotes.

Although three different nonsense codons can terminate chain elongation, UAA is most often used for this purpose. Three release factors (RF) assist the ribosome in recognizing nonsense codons and terminating translation. GTP hydrolysis is probably involved in termination.

tional proteins. Their role is essential because the cytoplasmic matrix is filled with nascent polypeptide chains and proteins. Under such conditions it is quite likely that new polypeptide chains often will fold improperly and aggregate to form nonfunctional complexes. Molecular chaperones suppress incorrect folding and may reverse any incorrect folding that has already taken place. They are so important that chaperones are present in all cells, prokaryotic and eukaryotic.

Several chaperones and cooperating proteins aid proper protein folding in bacteria. The process has been most studied in *Escherichia coli* and involves at least four chaperones—DnaK, DnaJ, GroEL, and GroES—and the stress protein GrpE. After a sufficient length of nascent polypeptide extends from the ribosome, DnaJ binds to the unfolded chain (**figure 12.18**). DnaK, which is complexed with ATP, then attaches to the polypeptide. These two chaperones prevent the polypeptide from folding improperly as it is synthesized. The ATP is hydrolyzed to ADP after DnaK binding, and this increases the affinity of DnaK for the un-

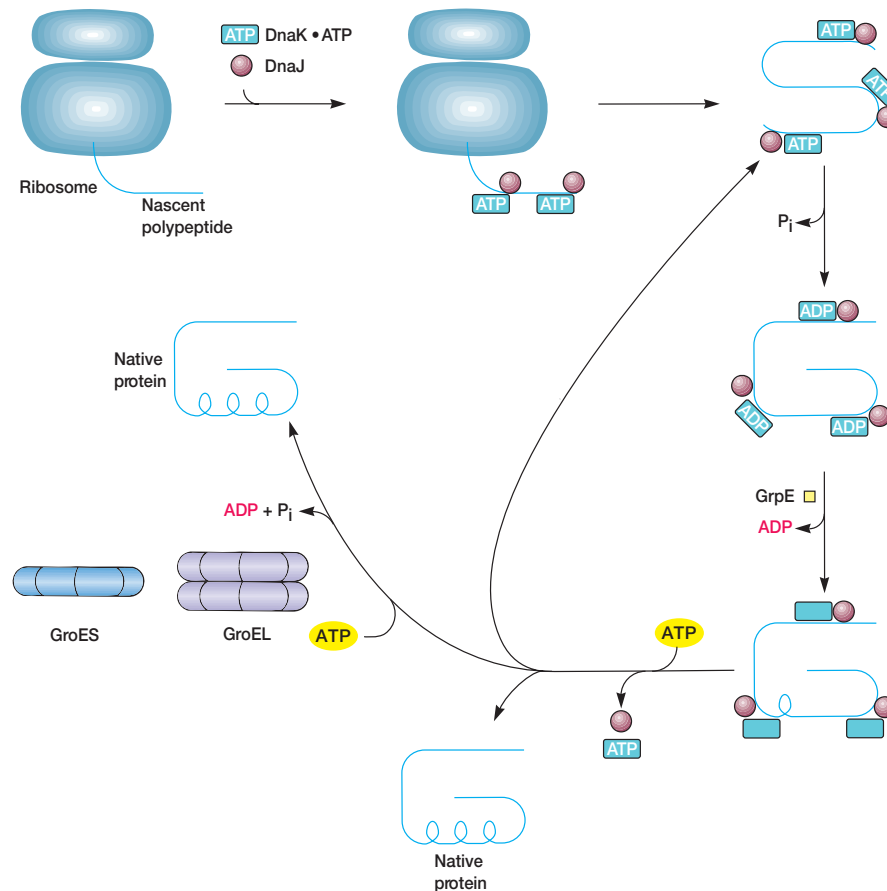


Figure 12.18 Chaperones and Polypeptide Folding. The involvement of bacterial chaperones in the proper folding of a newly synthesized polypeptide chain is depicted in this diagram. Three possible outcomes of a chaperone reaction cycle are shown. A native protein may result, the partially folded polypeptide may bind again to DnaK and DnaJ, or the polypeptide may be transferred to GroEL and GroES. See text for details.

folded polypeptide. When the polypeptide has been synthesized, the GrpE protein binds to the chaperone-polypeptide complex and causes DnaK to release ADP. Then ATP binds to DnaK and both DnaK and DnaJ dissociate from the polypeptide. The polypeptide has been folding during this sequence of events and may have reached its final native conformation. If it is still only partially folded, it can bind DnaJ and DnaK again and repeat the process. Often DnaK and DnaJ will transfer the polypeptide to the chaperones GroEL and GroES, where the final folding takes place. GroEL is a large, hollow barrel-shaped complex of 14 subunits arranged in two stacked rings (**figure 12.19**). GroES exists as a single ring of seven subunits and can bind to one or both ends of the GroEL cylinder. As with DnaK, ATP binding to GroEL and ATP hydrolysis change the chaperone's affinity for the folding polypeptide and regulate polypeptide binding and release (polypeptide release is ATP-dependent). GroES binds to GroEL and assists in its binding and release of the refolding polypeptide.

Chaperones were first discovered because they dramatically increased in concentration when cells were exposed to high tempera-

tures, metabolic poisons, and other stressful conditions. Thus many chaperones often are called **heat-shock proteins** or stress proteins. When an *E. coli* culture is switched from 30 to 42°C, the concentrations of some 20 different heat-shock proteins increase greatly within about 5 minutes. If the cells are exposed to a lethal temperature, the heat-shock proteins are still synthesized but most proteins are not. Thus chaperones protect the cell from thermal damage and other stresses as well as promote the proper folding of new polypeptides. For example, DnaK protects *E. coli* RNA polymerase from thermal inactivation in vitro. In addition, DnaK reactivates thermally inactivated RNA polymerase, especially if ATP, DnaJ, and GrpE are present. GroEL and GroES also protect intracellular proteins from aggregation. As one would expect, large quantities of chaperones are present in hyperthermophiles such as *Pyrodicticum occultum*, an archaeon that will grow at temperatures as high as 110°C. *Pyrodicticum* has a chaperone similar to the GroEL of *E. coli*. The chaperone hydrolyzes ATP most rapidly at 100°C and makes up almost 3/4 of the cell's soluble protein when *P. occultum* grows at 108°C. **Thermophilic and hyperthermophilic prokaryotes (pp. 126–27)**

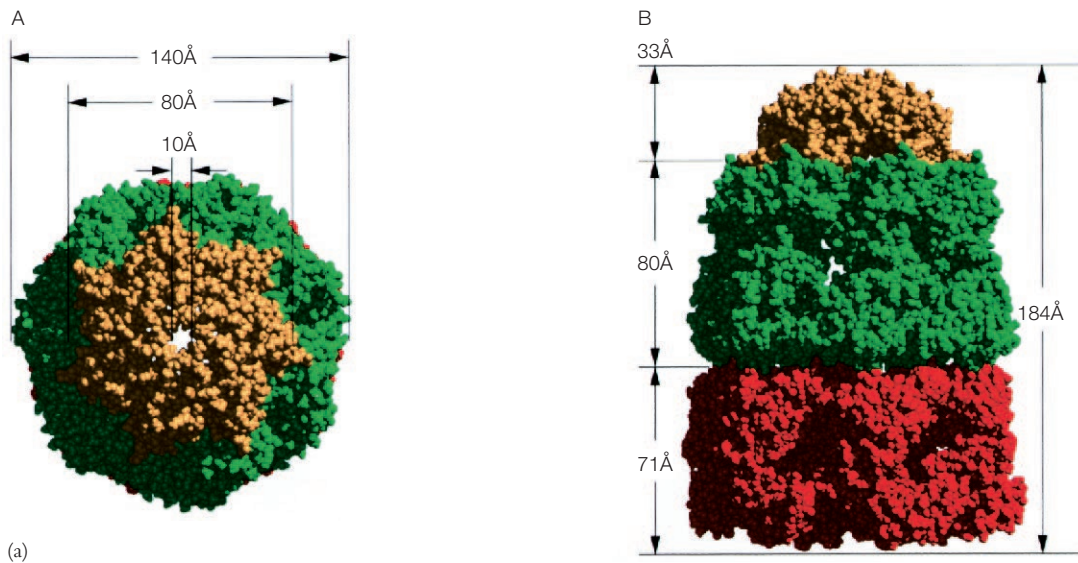


Figure 12.19 The GroEL-GroES Chaperone Complex. (a) Top and side views of the GroEL-GroES complex. The trans GroEL ring is red; the cis GroEL ring is green; GroES is gold. (b) A top view of the GroEL complex. Several domains have been colored to distinguish them from one another. Note the large central chamber in which a protein can fold.

Chaperones have other functions as well. They are particularly important in the transport of proteins across membranes. For example, in *E. coli* the chaperone SecB binds to the partially unfolded forms of many proteins and keeps them in an export-competent state until they are translocated across the plasma membrane. DnaK, DnaJ, and GroEL/GroES also can aid in protein translocation across membranes. Proteins destined for the periplasm or outer membrane (see pp. 58–60) are synthesized with the proper amino-terminal signal sequence. The signal sequence is a short stretch of amino acids that help direct the completed polypeptide to its final destination. Polypeptides associate with SecB and the chaperone then attaches to the membrane translocase. The polypeptides are transported through the membrane as ATP is hydrolyzed. When they enter the periplasm, the signal peptidase enzyme removes the signal sequence and the protein moves to its final location.

As already noted, the polypeptide folds into its final shape after synthesis, often with the aid of molecular chaperones. This

folding is possible because protein conformation is a direct function of the amino acid sequence (see appendix I). Recent research indicates that procaryotes and eucaryotes may differ with respect to the timing of protein folding. In terms of conformation, proteins are composed of compact, self-folding, structurally independent regions. These regions, normally around 100 to 300 amino acids in length, are called **domains**. Larger proteins such as immunoglobulins (see p. 734) may have two or more domains that are linked by less structured portions of the polypeptide chain. In eucaryotes, domains fold independently right after being synthesized by the ribosome. It appears that procaryotic polypeptides, in contrast, do not fold until after the complete chain has been synthesized. Only then do the individual domains fold. This difference in timing may account for the observation that chaperones seem to be more important in the folding of procaryotic proteins. Folding a whole polypeptide is more complex than folding one domain at a time and would require the aid of chaperones.

Protein Splicing

A further level of complexity in the formation of proteins has been discovered. Some microbial proteins are spliced after translation. In **protein splicing**, a part of the polypeptide is removed before the polypeptide folds into its final shape. Self-splicing proteins begin as larger precursor proteins composed of one or more internal intervening sequences called **inteins** flanked by external sequences or **exteins**, the N-exteins and C-exteins (**figure 12.20a**). Inteins, which sometimes are over 500 residues in length, are removed in an autocatalytic process involving a branched intermediate (**figure 12.20b**). Thus far, 10 or more self-splicing proteins have been discovered. Some examples are an ATPase in the yeast *Saccharomyces cerevisiae*, the recA protein of *Mycobacterium tuberculosis*, and DNA polymerase in *Pyrococcus*. The presence of self-splicing proteins in all three domains may mean that they are quite widespread and prevalent.

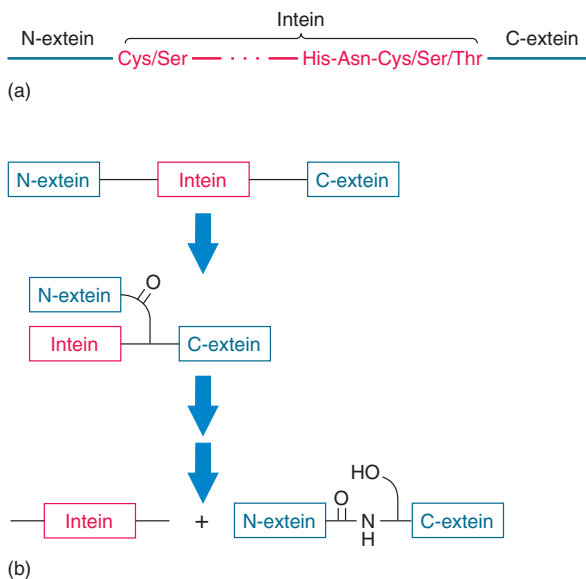


Figure 12.20 Protein Splicing. (a) A generalized illustration of intein structure. The amino acids that are commonly present at each end of the inteins are shown. Note that many are thiol- or hydroxyl-containing amino acids. (b) An overview of the proposed pattern or sequence of splicing. The precise mechanism is not yet known but presumably involves the hydroxyls or thiols located at each end of the intein.

1. In which direction are polypeptides synthesized? What is a polyribosome and why is it useful?
2. Briefly describe the structure of transfer RNA and relate this to its function. How are amino acids activated for protein synthesis, and why is the specificity of the aminoacyl-tRNA synthetase reaction so important?
3. What are translational and exit domains? Contrast procaryotic and eucaryotic ribosomes in terms of structure. What roles does ribosomal RNA have?
4. Describe the nature and function of the following: fMet-tRNA, initiator codon, IF-3, IF-2, IF-1, elongation cycle, peptidyl and aminoacyl sites, EF-Tu, EF-Ts, transpeptidation reaction, peptidyl transferase, translocation, EF-G or translocase, nonsense codon, and release factors.
5. What are molecular chaperones and heat-shock proteins? Describe their functions.

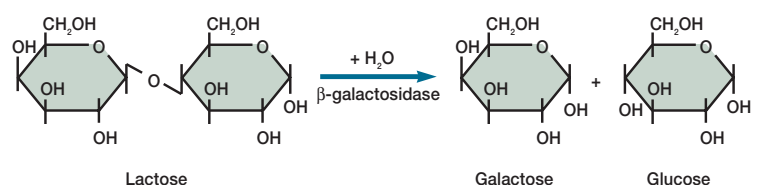
12.3 Regulation of mRNA Synthesis

The control of metabolism by regulation of enzyme activity is a fine-tuning mechanism: it acts rapidly to adjust metabolic activity from moment to moment. Microorganisms also are able to control the expression of their genome, although over longer intervals. For example, the *E. coli* chromosome can code for about 2,000 to 4,000 peptide chains, yet many fewer proteins are present in *E. coli* growing with glucose as its energy source. Regulation of gene expression serves to conserve energy and raw material, to maintain balance between the amounts of various cell proteins, and to adapt to long-term environmental change. Thus control of gene expression complements the regulation of enzyme activity. [Regulation of enzyme activity \(pp. 165–69\)](#)

Induction and Repression

The regulation of β -galactosidase synthesis has been intensively studied and serves as a primary example of how gene expression is controlled. This enzyme catalyzes the hydrolysis of the sugar lactose to glucose and galactose (**figure 12.21**). When *E. coli* grows with lactose as its carbon source, each cell contains about 3,000 β -galactosidase molecules, but has less than three molecules in the absence of lactose. The enzyme β -galactosidase is an **inducible enzyme**—that is, its level rises in the presence of a small molecule called an **inducer** (in this case the lactose derivative allolactose).

Figure 12.21 The β -Galactosidase Reaction.



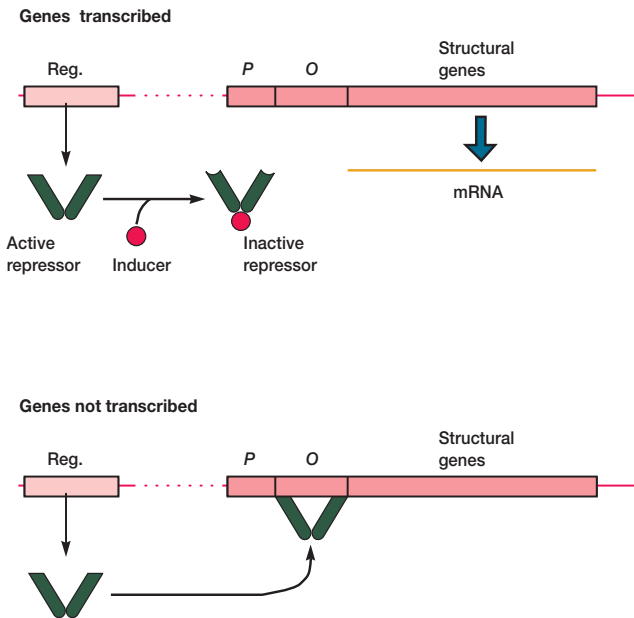


Figure 12.22 Gene Induction. The regulator gene, Reg., synthesizes an active repressor that binds to the operator, *O*, and blocks RNA polymerase binding to the promoter, *P*, unless the inducer inactivates it. In the presence of the inducer, the repressor protein is inactive and transcription occurs.

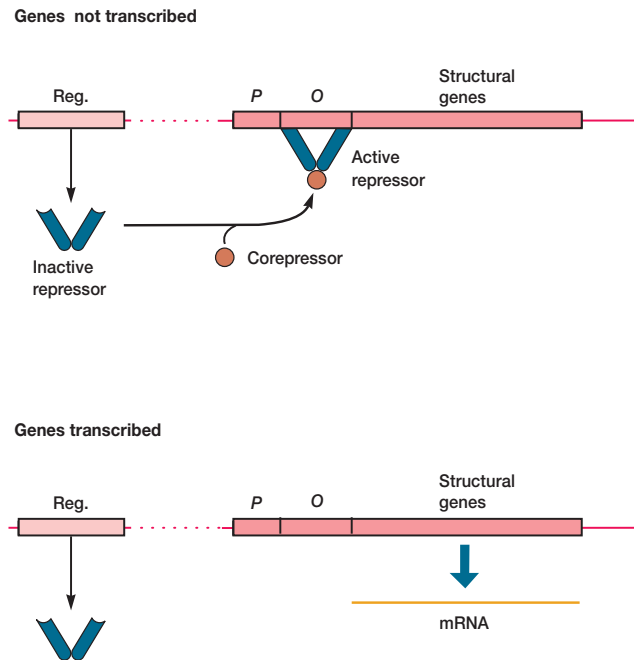


Figure 12.23 Gene Repression. The regulator gene, Reg., synthesizes an inactive repressor protein that must be activated by corepressor binding before it can bind to the operator, *O*, and block transcription. In the absence of the corepressor, the repressor is inactive and transcription occurs.

The genes for enzymes involved in the biosynthesis of amino acids and other substances often respond differently from genes coding for catabolic enzymes. An amino acid present in the surroundings may inhibit the formation of the enzymes responsible for its biosynthesis. This makes good sense because the microorganism will not need the biosynthetic enzymes for a particular substance if it is already available. Enzymes whose amount is reduced by the presence of an end product are **repressible enzymes**, and metabolites causing a decrease in the concentrations of repressible enzymes are **corepressors**. Generally, repressible enzymes are necessary for synthesis and always are present unless the end product of their pathway is available. Inducible enzymes, in contrast, are required only when their substrate is available; they are missing in the absence of the inducer.

Although variations in enzyme levels could be due to changes in the rates of enzyme degradation, most enzymes are relatively stable in growing bacteria. Induction and repression result principally from changes in the rate of transcription. When *E. coli* is growing in the absence of lactose, it often lacks mRNA molecules coding for the synthesis of β -galactosidase. In the presence of lactose, however, each cell has 35 to 50 β -galactosidase mRNA molecules. The synthesis of mRNA is dramatically influenced by the presence of lactose. [DNA transcription mechanism \(pp. 261–64\)](#)

Negative Control

A controlling factor can either inhibit or activate transcription. Although the responses to the presence of metabolites are different, both induction and repression are forms of **negative control**: mRNA synthesis proceeds more rapidly in the absence of the active controlling factor.

The rate of mRNA synthesis is controlled by special **repressor proteins** that are synthesized under the direction of regulator genes. The repressor binds to a specific site on DNA called the **operator**. The importance of regulator genes and repressors is demonstrated by mutationally inactivating a regulator gene to form a **constitutive mutant**. A constitutive mutant produces the enzymes in question whether or not they are needed. Thus inactivation of repressor proteins blocks the regulation of transcription. [Gene structure \(pp. 241–44\)](#)

Repressors must exist in both active and inactive forms because transcription would never occur if they were always active. In inducible systems the regulator gene directs the synthesis of an active repressor. The inducer stimulates transcription by reversibly binding to the repressor and causing it to change to an inactive shape (**figure 12.22**). Just the opposite takes place in a system controlled by repression (**figure 12.23**). The repressor protein initially is an inactive form called an **aporepressor** and becomes an active repressor only when the corepressor binds to it. The corepressor inhibits transcription by activating the aporepressor.

Box 12.2

The Discovery of Gene Regulation

The ability of microorganisms to adapt to their environments by adjusting enzyme levels was first discovered by Emil Duclaux, a colleague of Louis Pasteur. He found that the fungus *Aspergillus niger* would produce the enzyme that hydrolyzes sucrose (invertase) only when grown in the presence of sucrose. In 1900 F. Diernt found that yeast contained the enzymes for galactose metabolism only when grown with lactose or galactose and would lose these enzymes upon transfer to a glucose medium. Such a response made sense because the yeast cells would not need enzymes for galactose metabolism when using glucose as its carbon and energy source. Further examples of adaptation were discovered and by the 1930s H. Karström could divide enzymes into two classes: (1) adaptive enzymes that are formed only in the presence of their substrates, and (2) constitutive enzymes that are always present. It was originally thought that enzymes might be formed from inactive precursors and that the presence of the substrate simply shifted the equilibrium between precursor and enzyme toward enzyme formation.

In 1942 Jacques Monod, working at the Pasteur Institute in Paris, began a study of adaptation in the bacterium *E. coli*. It was already known that the enzyme β -galactosidase, which hydrolyzes the sugar lactose to glucose and galactose, was present only when *E. coli* was grown in the presence of lactose. Monod discovered that nonmetabolizable analogues of β -galactosides, such as thiomethylgalactoside, also could induce enzyme production. This discovery made it possible to study induction in cells growing on carbon and energy sources other than lactose so that the growth rate and inducer concentration would not depend on the lactose supply. He next demonstrated that induction involved the synthesis of new enzyme, not just the conversion of already available precursor. Monod accomplished this by making *E. coli* proteins radioactive with ^{35}S , then transferring the labeled bacteria to nonradioactive medium and adding inducer. The newly formed

β -galactosidase was nonradioactive and must have been synthesized after addition of inducer.

A study of the genetics of lactose induction in *E. coli* was begun by Joshua Lederberg a few years after Monod had started his work. Lederberg isolated not only mutants lacking β -galactosidase but also a constitutive mutant in which synthesis of the enzyme proceeded in the absence of an inducer (LacI^-). During bacterial conjugation (see section 13.4), genes from the donor bacterium enter the recipient to temporarily form an organism with two copies of those genes provided by the donor. When Arthur B. Pardee, François Jacob, and Monod transferred the gene for inducibility to a constitutive recipient not sensitive to inducers, the newly acquired gene made the recipient bacterium sensitive to inducer again. This functional gene was not a part of the recipient's chromosome. Thus the special gene directed the synthesis of a cytoplasmic product that inhibited the formation of β -galactosidase in the absence of the inducer. In 1961 Jacob and Monod named this special product the repressor and suggested that it was a protein. They further proposed that the repressor protein exerted its effects by binding to the operator, a special site next to the structural genes. They provided genetic evidence for their hypothesis. The name operon was given to the complex of the operator and the genes it controlled. Several years later in 1967, Walter Gilbert and Benno Müller-Hill managed to isolate the lac repressor and show that it was indeed a protein and did bind to a specific site in the lac operon.

The existence of repression was discovered by Monod and G. Cohen-Bazire in 1953 when they found that the presence of the amino acid tryptophan would repress the synthesis of tryptophan synthetase, the final enzyme in the pathway for tryptophan biosynthesis. Subsequent research in many laboratories showed that induction and repression were operating by quite similar mechanisms, each involving repressor proteins that bound to operators on the genome.

The synthesis of several proteins is often regulated by a single repressor. The **structural genes**, or genes coding for a polypeptide, are simply lined up together on the DNA, and a single mRNA carries all the messages. The sequence of bases coding for one or more polypeptides, together with the operator controlling its expression, is called an **operon**. This arrangement is of great advantage to the bacterium because coordinated control of the synthesis of several metabolically related enzymes (or other proteins) can be achieved.

The Lactose Operon

The best-studied negative control system is the lactose operon of *E. coli*. The lactose or *lac* operon contains three structural genes and is controlled by the *lac* repressor (figure 12.24). One gene codes for β -galactosidase; a second gene directs the synthesis of β -galactoside permease, the protein responsible for lactose uptake. The third gene codes for the enzyme β -galactoside transacetylase, whose function still is uncertain. The presence of the first two genes in the same operon ensures that the rates of lactose uptake and breakdown will vary together (Box 12.2).

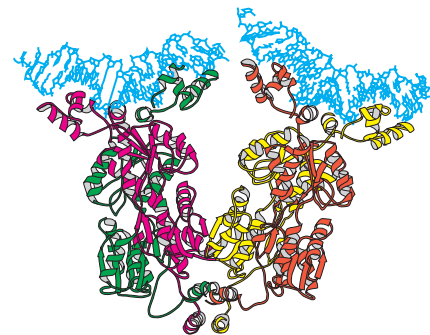


Figure 12.24 Lactose Repressor Binding to DNA. The *lac* repressor-DNA complex is shown here. The repressor dimer binds to two stretches of DNA (blue) by specialized N-terminal headpiece subdomains that fit in the major groove.

The *lac* operon has three operators. The *lac* repressor protein finds an operator in a two-step process. First, the repressor binds to a DNA molecule, then rapidly slides along the DNA until it reaches an operator and stops. A portion of the repressor fits into

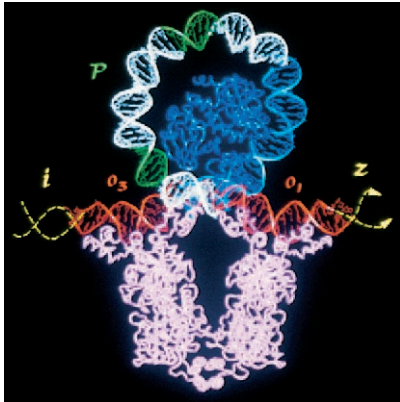


Figure 12.25 Repressor and CAP Bound to the *lac* Operon. The *lac* repressor is in violet, operators in red, promoter in green, and CAP (catabolite activator protein) in blue. RNA polymerase access to the promoter in the loop of DNA is hindered in this complex and transcription cannot begin.

the major groove of operator-site DNA by special N-terminal subdomains. The shape of the repressor protein is ideally suited for specific binding to the DNA double helix.

How does the repressor inhibit transcription? The promoter to which RNA polymerase binds (*see p. 242–44*) is located next to the operator. The repressor may bind simultaneously to more than one operator and bend the DNA segment that contains the promoter (**figure 12.25**). The bent promoter may not allow proper RNA polymerase binding or may not be able to initiate transcription after polymerase binding. Even if the polymerase is bound to the promoter, it is stored there and does not begin transcription until the repressor leaves the operator. A repressor does not affect the actual rate of transcription once it has begun.

Positive Control

The preceding section shows that operons can be under negative control, resulting in induction and repression. In contrast, some operons function only in the presence of a controlling factor—that is, they are under **positive operon control**. The *lac* operon is under positive control as well as negative control—that is, it is under dual control.

Lac operon function is regulated by the **catabolite activator protein (CAP)** or **cyclic AMP receptor protein (CRP)** and the small cyclic nucleotide 3', 5' -**cyclic adenosine monophosphate (cAMP; figure 12.26)**, as well as by the *lac* repressor protein. The *lac* promoter contains a CAP site to which CAP must bind before RNA polymerase can attach to the promoter and begin transcription (**figure 12.27**). The catabolite activator protein is able to bind to the CAP site only when complexed with cAMP. Upon binding, CAP bends the DNA about 90° within two helical turns (**figure 12.25** and **figure 12.28**). Interaction of CAP with RNA polymerase stimulates transcription. This positive control system makes *lac* operon activity dependent on the presence of cAMP as well as on that of lactose.

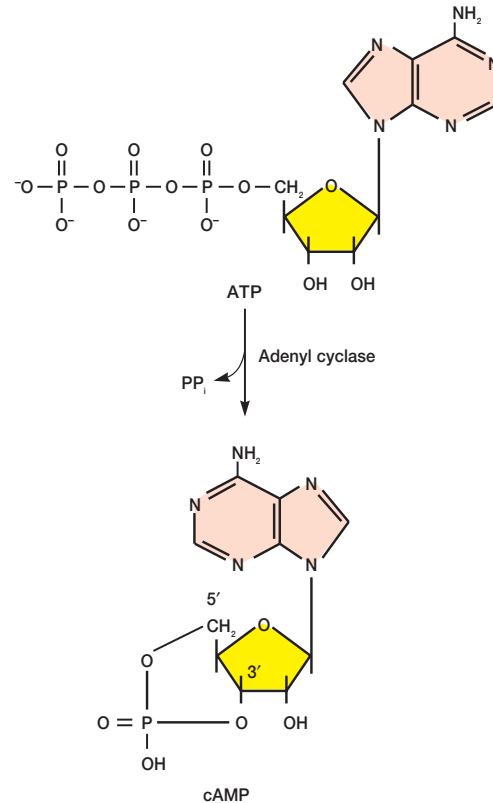


Figure 12.26 Cyclic Adenosine Monophosphate (cAMP). The phosphate group extends between the 3' and 5' hydroxyls of the ribose sugar. The enzyme adenylyl cyclase forms cAMP from ATP.

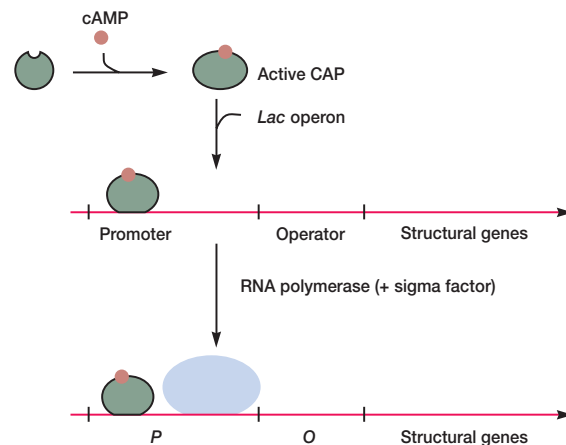


Figure 12.27 Positive Control of the *Lac* Operon. When cyclic AMP is absent or present at a low level, the CAP protein remains inactive and does not bind to the promoter. In this situation RNA polymerase also does not bind to the promoter and transcribe the operon's genes.

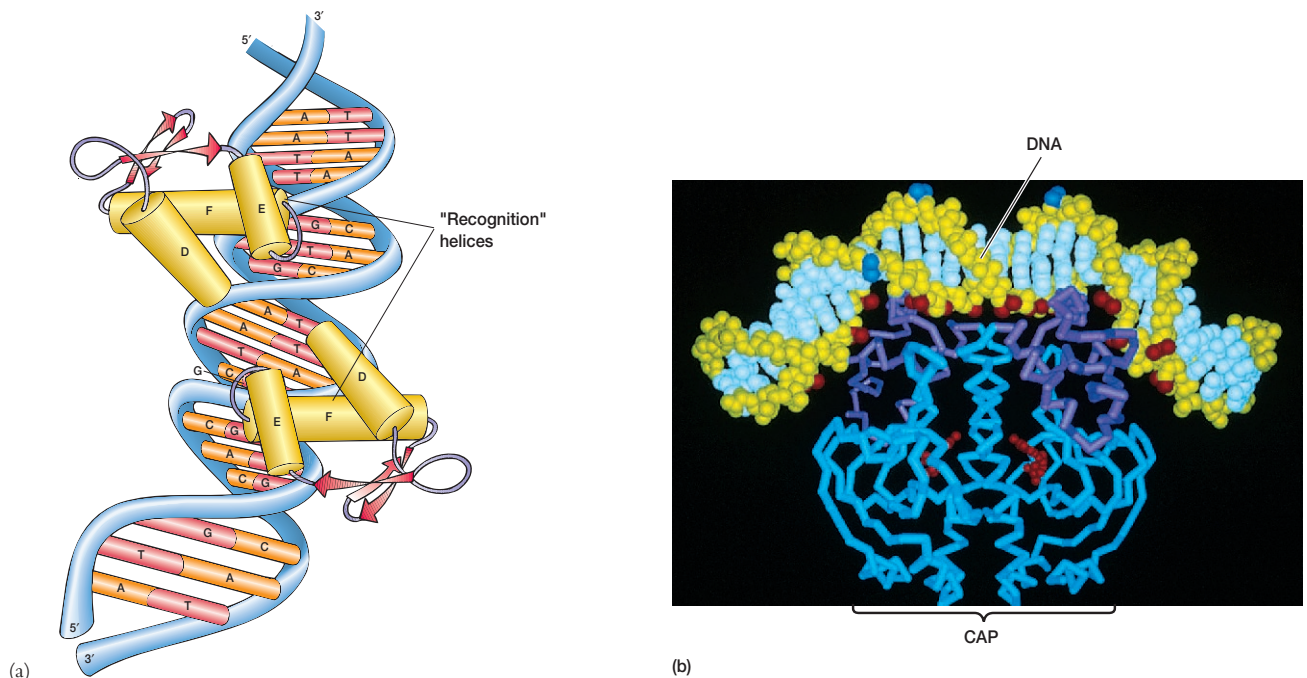


Figure 12.28 CAP Structure and DNA Binding. (a) The CAP dimer binding to DNA at the *lac* operon promoter. The recognition helices fit into two adjacent major grooves on the double helix. (b) A model of the *E. coli* CAP-DNA complex derived from crystal structure studies. The cAMP binding domain is in blue and the DNA binding domain, in purple. The cAMP molecules bound to CAP are in red. Note that the DNA is bent by 90° when complexed with CAP.

12.4 Attenuation

Bacteria can regulate transcription in other ways, as may be seen in the tryptophan operon of *E. coli*. The tryptophan operon contains structural genes for five enzymes in this amino acid's biosynthetic pathway. As might be expected, the operon is under the control of a repressor protein coded for by the *trpR* gene (*trp* stands for tryptophan), and excess tryptophan inhibits transcription of operon genes by acting as a corepressor and activating the repressor protein. Although the operon is regulated mainly by repression, the continuation of transcription also is controlled. That is, there are two decision points involved in transcriptional control, the initiation of transcription and the continuation of transcription past the attenuator region.

A **leader region** lies between the operator and the first structural gene in the operon, the *trpE* gene, and is responsible for controlling the continuation of transcription after the RNA polymerase has bound to the promoter (figure 12.29a). The leader region contains an **attenuator** and a sequence that codes for the synthesis of a leader peptide. The attenuator is a rho-independent termination site (p. 263) with a short GC-rich segment followed by a sequence of eight U residues. The four stretches marked off in figure 12.29a have complementary base sequences and can base pair with each other to form hairpin loops. In the absence of a ribosome, mRNA segments one and two pair to form a hairpin,

while segments three and four generate a second loop next to the poly(U) sequence (figure 12.29b). The hairpin formed by segments three and four plus the poly(U) sequence will terminate transcription. If segment one is prevented from base pairing with segment two, segment two is free to associate with segment three. As a result segment four remains single stranded (figure 12.29c) and cannot serve as a terminator for transcription. It is important to note that the sequence coding for the leader peptide contains two adjacent codons that code for the amino acid tryptophan. Thus the complete peptide can be made only when there is an adequate supply of tryptophan. Since the leader peptide has not been detected, it must be degraded immediately after synthesis.

Ribosome behavior during translation of the mRNA regulates RNA polymerase activity as it transcribes the leader region. This is possible because translation and transcription are tightly coupled. When the active repressor is absent, RNA polymerase binds to the promoter and moves down the leader synthesizing mRNA. If there is no translation of the mRNA after the RNA polymerase has begun copying the leader region, segments three and four form a hairpin loop, and transcription terminates before the polymerase reaches the *trpE* gene (figure 12.30a). When tryptophan is present, there is sufficient tryptophanyl-tRNA for protein synthesis. Therefore the ribosome will synthesize the leader peptide and continue moving along the mRNA until it reaches a UGA stop codon (see section 12.2) lying between

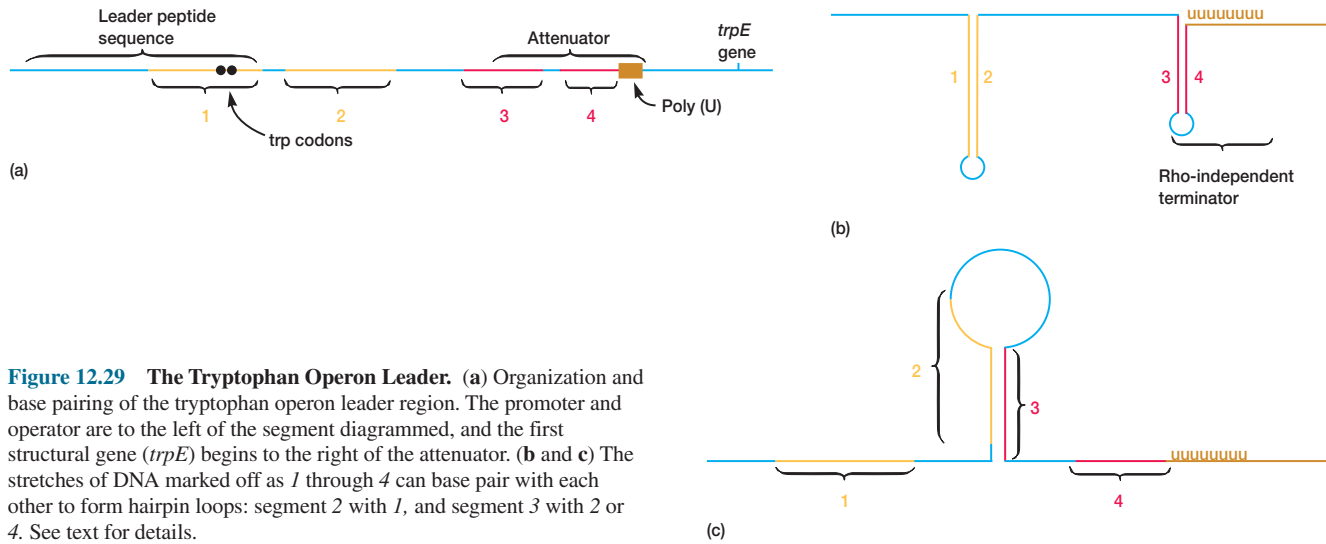


Figure 12.29 The Tryptophan Operon Leader. (a) Organization and base pairing of the tryptophan operon leader region. The promoter and operator are to the left of the segment diagrammed, and the first structural gene (*trpE*) begins to the right of the attenuator. (b) and (c) The stretches of DNA marked off as 1 through 4 can base pair with each other to form hairpin loops: segment 2 with 1, and segment 3 with 2 or 4. See text for details.

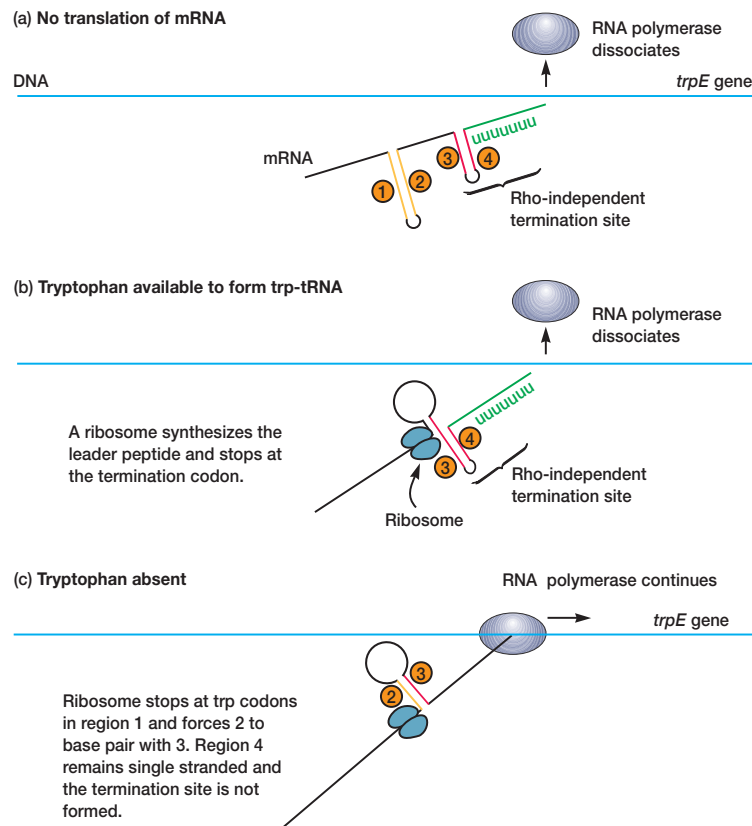


Figure 12.30 Attenuation Control. The control of tryptophan operon function by attenuation. See text for details.

segments one and two. The ribosome halts at this codon and projects into segment two far enough to prevent it from pairing properly with segment three (figure 12.30*b*). Segments three and four form a hairpin loop, and the RNA polymerase terminates at the attenuator just as if no translation had taken place. If tryptophan is lacking, the ribosome will stop at the two adjacent tryptophan codons in the leader peptide sequence and prevent segment one from base pairing with segment two, because the tryptophan codons are located within segment one (figures 12.29*a* and 12.30*c*). If this happens while the RNA polymerase is still transcribing the leader region, segments two and three associate before segment four has been synthesized. Therefore segment four will remain single stranded and the terminator hairpin will not form. Consequently, when tryptophan is absent, the RNA polymerase continues on and transcribes tryptophan operon genes. Control of the continuation of transcription by a specific aminoacyl-tRNA is called **attenuation**.

Attenuation's usefulness is apparent. If the bacterium is deficient in an amino acid other than tryptophan, protein synthesis will slow and tryptophanyl-tRNA will accumulate. Transcription of the tryptophan operon will be inhibited by attenuation. When the bacterium begins to synthesize protein rapidly, tryptophan may be scarce and the concentration of tryptophanyl-tRNA may be low. This would reduce attenuation activity and stimulate operon transcription, resulting in larger quantities of the tryptophan biosynthetic enzymes. Acting together, repression and attenuation can coordinate the rate of synthesis of amino acid biosynthetic enzymes with the availability of amino acid end products and with the overall rate of protein synthesis. When tryptophan is present at high concentrations, any RNA polymerases not blocked by the activated repressor protein probably will not get past the attenuator sequence. Repression decreases transcription about seventyfold and attenuation slows it another eight- to tenfold; when both mechanisms operate together, transcription can be slowed about 600-fold.

Attenuation seems important in the regulation of several amino acid biosynthetic pathways. At least five other operons have leader peptide sequences that resemble the tryptophan system in organization. For example, the leader peptide sequence of the histidine operon codes for seven histidines in a row and is followed by an attenuator that is a terminator sequence.

12.5 Global Regulatory Systems

Thus far, we have been considering the function of isolated operons. However, bacteria must respond rapidly to a wide variety of changing environmental conditions and be able to cope with such things as nutrient deprivation, desiccation, and major temperature fluctuations. They also have to compete successfully with other organisms for scarce nutrients and use these nutrients efficiently. These challenges require a regulatory system that can rapidly control many operons at the same time. Such regulatory systems that affect many genes and pathways simultaneously are called **global regulatory systems**. There are many examples of these multigene global systems. Catabolite repression in enteric bacteria and sporulation in *Bacillus subtilis* will be discussed shortly. Two other pre-

viously discussed global systems are the SOS response (*see p. 255*) and the production of heat-shock proteins (*p. 273*).

Although it is usually possible to regulate all the genes of a metabolic pathway in a single operon, there are good reasons for more complex global systems. Some processes involve too many genes to be accommodated in a single operon. For example, the machinery required for protein synthesis is composed of 150 or more gene products, and coordination requires a regulatory network that controls many separate operons. Sometimes two levels of regulation are required because individual operons must be controlled independently and also cooperate with other operons. Regulation of sugar catabolism in *E. coli* is a good example. *E. coli* uses glucose when it is available; in such a case, operons for other catabolic pathways are repressed. If glucose is unavailable and another nutrient is present, the appropriate operon is activated.

Global regulation can be accomplished by several different mechanisms. A protein repressor or activator may affect several operons simultaneously. A sigma factor may cause RNA polymerase to recognize and transcribe an array of different operons with similar promoters. Sometimes a nonprotein regulator such as the nucleotide guanosine tetraphosphate controls several operons.

Global regulatory systems are so complex that a specialized nomenclature is used to describe the various kinds. Perhaps the most basic type is the **regulon**. A regulon is a collection of genes or operons that is controlled by a common regulatory protein. Usually the operons are associated with a single pathway or function (e.g., the production of heat-shock proteins or the catabolism of glycerol). A somewhat more complex situation is seen with a **modulon**. This is an operon network under the control of a common global regulatory protein, but whose constituent operons also are controlled separately by their own regulators. A good example of a modulon is catabolite repression. The most complex global systems are referred to as **stimulons**. A stimulon is a regulatory system in which all operons respond together in a coordinated way to an environmental stimulus. It may contain several regulons and modulons, and some of these may not share regulatory proteins. The genes involved in a response to phosphate limitation are scattered among several regulons and are part of one stimulon.

We will now briefly consider three examples of global regulation. First we will discuss catabolite repression and the use of positive operon control. Then an introduction to regulation by sigma factors and the induction of sporulation will follow. Finally, the regulation of porin protein synthesis by antisense RNA will be described.

Catabolite Repression

If *E. coli* grows in a medium containing both glucose and lactose, it uses glucose preferentially until the sugar is exhausted. Then after a short lag, growth resumes with lactose as the carbon source (figure 12.31). This biphasic growth pattern or response is called **diauxic growth**. The cause of diauxic growth or diauxie is complex and not completely understood, but **catabolite repression** or the glucose effect probably plays a part. The enzymes for glucose catabolism are constitutive and unaffected by CAP activity. When the bacterium is given glucose, the cAMP level drops, resulting in

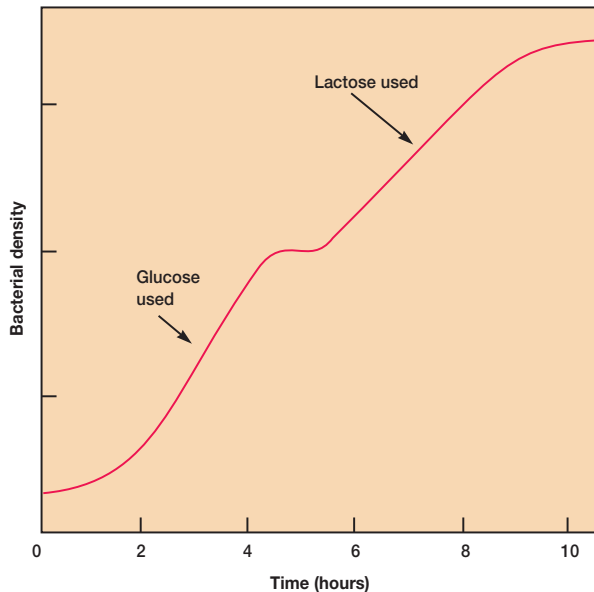


Figure 12.31 Diauxic Growth. The diauxic growth curve of *E. coli* grown with a mixture of glucose and lactose. Glucose is first used, then lactose. A short lag in growth is present while the bacteria synthesize the enzymes needed for lactose use.

deactivation of the catabolite activator protein and inhibition of lac operon expression. The decrease in cAMP may be due to the effect of the phosphoenolpyruvate:phosphotransferase system (PTS) on the activity of adenyl cyclase, the enzyme that synthesizes cAMP. Enzyme III of the PTS donates a phosphate to glucose during its transport; therefore, it enters the cell as glucose 6-phosphate. The phosphorylated form of enzyme III also activates adenyl cyclase. If glucose is being rapidly transported by PTS, the amount of phosphorylated enzyme III is low and the adenyl cyclase is less active, so the cAMP level drops. At least one other mechanism is involved in diauxic growth. When the PTS is actively transporting glucose into the cell, nonphosphorylated enzyme III is more prevalent. Nonphosphorylated enzyme III binds to the lactose permease and allosterically inhibits it, thus blocking lactose uptake. [The phosphoenolpyruvate: phosphotransferase system \(pp. 103–4\)](#)

Whatever the precise mechanism, such control is of considerable advantage to the bacterium. It will use the most easily catabolized sugar (glucose) first rather than synthesize the enzymes necessary for another carbon and energy source. These control mechanisms are present in a variety of bacteria and metabolic pathways.

Regulation by Sigma Factors and Control of Sporulation

Although the RNA polymerase core enzyme can transcribe any gene to produce a messenger RNA copy, it needs the assistance of a sigma factor to bind the promoter and initiate transcription (see p. 262). This provides an excellent means of regulating gene expression. When a complex process requires a radical change in

transcription, or the synthesis of several gene products in a precisely timed sequence, it may be regulated by a series of sigma factors. Each sigma factor enables the RNA polymerase core enzyme to recognize a specific set of promoters and transcribe only those genes. Substitution of the sigma factor immediately changes gene expression. Bacterial viruses often use sigma factors to control mRNA synthesis during their life cycle (*see chapter 17*). This regulatory mechanism also is common among both gram-negative and gram-positive bacteria. For example, *Escherichia coli* synthesizes several sigma factors. Under normal conditions the sigma factor σ^{70} directs RNA polymerase activity. (The superscript letter or number indicates the function or size of the sigma factor; 70 stands for 70,000 Da.) When flagella and chemotactic proteins are needed, *E. coli* produces σ^F (σ^{28}). If the temperature rises too high, σ^H (σ^{32}) appears and stimulates the formation of around 17 heat-shock proteins to protect the cell from thermal destruction. As would be expected, the promoters recognized by each sigma factor differ characteristically in sequence at the -10 and -35 positions (*see pp. 242–44*).

One of the best-studied examples of gene regulation by sigma factors is the control of sporulation in the gram-positive *Bacillus subtilis*. When *B. subtilis* is deprived of nutrients, it will form endospores in a complex developmental process lasting about 8 hours. [The bacterial endospore \(pp. 68–71\)](#)

Normally the *B. subtilis* RNA polymerase uses sigma factor σ^A (σ^{43}) to recognize genes. Environmental signals such as nutrient deprivation stimulate a kinase (Kin A or Kin B) to catalyze the phosphorylation of the Spo0F protein (**figure 12.32**). Spo0F transfers the phosphate to Spo0B, which in turn phosphorylates Spo0A. Phosphorylated Spo0A has several effects. It binds to a promoter and represses the expression of the *abrB* gene. *abrB* codes for a protein that inhibits many genes not needed during growth with excess nutrients (e.g., at least three sporulation genes). Phosphorylated Spo0A also activates the production of two sigma factors: an active σ^F and an inactive pro- σ^E . Sporulation begins when σ^F partially replaces σ^A in the forespore. The RNA polymerase then transcribes sporulation genes as well as vegetative genes. One of these early sporulation genes codes for another sigma factor, σ^G , that causes RNA polymerase to transcribe late sporulation genes in the forespore. At this point the pro- σ^E protein is activated by cleavage to yield σ^E , which then stimulates the transcription of the gene for pro- σ^K . Pro- σ^K is activated by a protease to yield σ^K and trigger the transcription of late genes in the mother cell. In summary, sporulation is regulated by two cascades of sigma factors, one in the forespore and the other in the mother cell. Each cascade influences the other through a series of signals so that the whole complex developmental process is properly coordinated.

Antisense RNA and the Control of Porin Proteins

Microbiologists have known for many years that gene expression can be controlled by both regulatory proteins (e.g., repressor proteins and CAP) and aminoacyl-tRNA (attenuation). More recently it has been discovered that the activity of some genes is controlled by a special type of small regulatory RNA molecule. The regula-

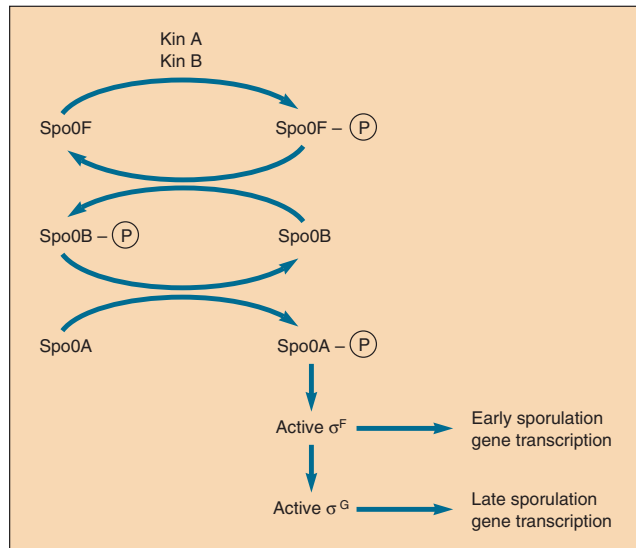


Figure 12.32 Initiation of Sporulation in *Bacillus subtilis*. A simplified diagram of the initial steps in triggering sporulation. The activation of kinases A and B begins a two-component phosphorelay system that activates the transcription regulator Spo0A. See text for more detail.

tory RNA, called **antisense RNA**, has a base sequence complementary to a segment of another RNA molecule and specifically binds to the target RNA. Antisense RNA binding can block DNA replication, mRNA synthesis, or translation. The genes coding for these RNAs are sometimes called antisense genes.

This mode of regulation appears to be widespread among viruses and bacteria. Examples are the regulation of plasmid replication and Tn10 transposition, osmoregulation of porin protein expression, regulation of λ phage reproduction, and the autoregulation of cAMP-receptor protein synthesis. Antisense RNA regulation has not yet been demonstrated in eucaryotic cells, although there is evidence that it may exist. It is possible that antisense RNAs bind with some eucaryotic mRNAs and stimulate their degradation. [Plasmids and transposons \(pp. 294–302\)](#)

The regulation of *E. coli* outer membrane porin proteins provides an example of control by antisense RNA. The outer membrane contains channels made of porin proteins (see p. 60). The two most important porins in *E. coli* are the OmpF and OmpC proteins. OmpC pores are slightly smaller and are made when the bacterium grows at high osmotic pressures. It is the dominant porin in *E. coli* from the intestinal tract. This makes sense because the smaller pores would exclude many of the toxic molecules present in the intestine. The larger OmpF pores are favored when *E. coli* grows in a dilute environment, and they allow solutes to diffuse into the cell more readily.

The *ompF* and *ompC* genes are partly regulated by a special OmpR protein that represses the *ompF* gene and activates *ompC*. In addition, the *micF* gene produces a 174-nucleotide-long antisense micF RNA that blocks *ompF* action (mic stands for mRNA-

interfering complementary RNA). The micF RNA is complementary to *ompF* at the translation initiation site. It complexes with *ompF* mRNA and represses translation. The *micF* gene is activated by conditions such as high osmotic pressure or the presence of some toxic materials that favor *ompC* expression. This helps ensure that OmpF protein is not produced at the same time as OmpC protein.

The fact that antisense RNA can bind specifically to mRNA and block its activity has great practical implications. Antisense RNA is already a valuable research tool. Suppose one desires to study the action of a particular gene. An antisense RNA that will bind to the gene's mRNA can be constructed and introduced into the cell, thus blocking gene expression. Changes in the cell are then observed. It also is possible to use the same approach with short strands of antisense DNA that bind to mRNA.

Antisense RNA and DNA may well be effective against a variety of cancers and infectious diseases. Promising preliminary results have been obtained using antisense oligonucleotides directed against *Trypanosoma brucei brucei* (the cause of African sleeping sickness), herpesviruses, the HIV virus, tumor viruses such as the RSV and polyoma viruses, ovarian cancer, cytomegalovirus infections, Crohn's disease, and chronic myelogenous leukemia. Although much further research is needed to determine the medical potential of these molecules, they may prove invaluable in the treatment of many diseases.

1. What are induction and repression and why are they useful? Define inducer, corepressor, repressor protein, aporepressor, regulator gene, negative control, constitutive mutant, operator, structural gene, and operon. Describe how the *lac* operon is regulated.
2. Define positive control, dual control, and catabolite activator protein. How is the *lac* operon controlled positively?
3. Define attenuation and describe how it works in terms of a labeled diagram, such as that provided in figure 12.30. What are the functions of the leader region and the attenuator in attenuation?
4. What are global regulatory systems and why are they necessary? Briefly describe regulons, modulons, and stimulons.
5. What is diauxic growth and how does it result from catabolite repression?
6. Briefly describe how sigma factors can be used to control gene expression. Describe the regulation of sporulation by sigma factors.
7. What is antisense RNA? How does it regulate gene expression?

12.6 Two-Component Phosphorelay Systems

A **two-component phosphorelay system** is a signal transduction system that uses the transfer of phosphoryl groups to control gene transcription and protein activity. It has two major components: a sensor kinase and a response regulator. There are many phosphorelay systems; two good examples are the systems that control sporulation and chemotaxis.

In the sporulation regulation system, kin A is a sensor kinase. It serves as a transmitter that phosphorylates itself (autophosphorylation) on a special histidine residue in response to environmental signals. The Spo0F acts as a receiver and catalyzes the

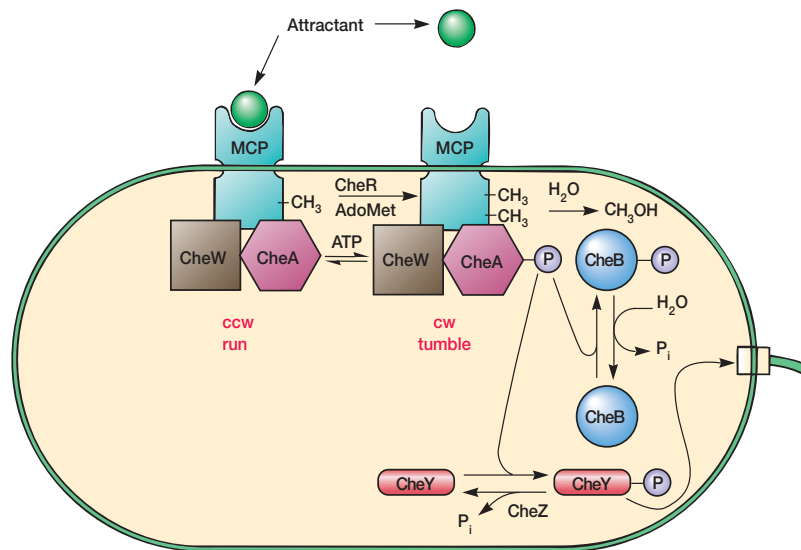


Figure 12.33 The Mechanism of Chemotaxis in *Escherichia coli*. The chemotaxis system is designed to control counterclockwise (ccw) and clockwise (cw) flagellar rotation so that *E. coli* moves up an attractant gradient by a sequence of runs and tumbles. See the text for a description of the process.

transfer of the phosphoryl group from kin A to a special aspartic acid residue on its surface; Spo0F then donates the phosphoryl group to a histidine on Spo0B. Spo0A is a response regulator. It has a receiver domain aspartate and picks up the phosphoryl group from Spo0B to become an active transcription regulator.

Control of sporulation by sigma factors (p. 282)

Chemotaxis is controlled by a well-studied phosphorelay regulatory system. As we have seen previously, procaryotes sense various chemicals in their environment when these substances bind to chemoreceptors called **methyl-accepting chemotaxis proteins (MCPs)**. The MCPs can influence flagellar rotation in such a way that the organisms swim toward attractants and away from repellants. This response is regulated by a complex system in which the CheA protein serves as a sensor kinase and the CheY protein is the response regulator. [Chemotaxis \(pp. 66–68\)](#)

The MCP chemoreceptors are buried in the plasma membrane with major parts exposed on both sides. The periplasmic side of each MCP has a binding site for one or more attractant molecules and may also bind repellants. Although attractants often bind directly to the MCP, in some cases they may attach to special periplasmic binding proteins, which then interact with the MCPs. The cytoplasmic side of an MCP interacts with two proteins (**figure 12.33**). The CheW protein binds to MCPs and helps attach the CheA protein. The full complex is composed of an MCP dimer, two CheW monomers, and a CheA dimer. When the MCP is not bound to an attractant, it stimulates CheA to phosphorylate itself using ATP, a process called autophosphorylation. CheA autophosphorylation is inhibited when the attractant is bound to its MCP. Phosphorylated CheA can donate its phosphate to one of two receptor proteins, CheY or CheB. If CheY is phosphorylated by CheA, it changes to an active conformation, moves to the flagellum, and in-

teracts with the switch protein (Fli M) at its base (*see Figure 3.36*). This causes the flagellum to rotate clockwise. Thus a decrease in attractant level promotes clockwise rotation and tumbling. The phosphate is removed from CheY in about 10 seconds in a process aided by the protein CheZ. The short lifetime of phosphorylated CheY means that the bacterium is very responsive to changes in attractant concentration. It will not be stuck in the tumble mode for too long a time when the attractant level changes. When no attractants or repellants are present, the system maintains intermediate concentrations of CheA phosphate and CheY phosphate. This produces a normal run-tumble swimming pattern.

The *E. coli* cell must ignore past stimulus responses so that it can compare the most recent attractant or repellent concentration with the immediately previous one and respond to any changes. This means that it must be able to adapt to a concentration change in order to detect still further changes. That is, it must have a short-term memory with a retention time of only seconds. This adaptation is accomplished by methylation of the MCP receptors. The cytoplasmic portion or domain of MCP molecules usually has about four or five methylation sites containing special glutamic acid residues. Methyls can be added to these glutamic acid carboxyl groups using S-adenosylmethionine as the methylating agent. The reaction is catalyzed by the CheR protein and occurs at a fairly steady rate regardless of the attractant level. Methyl groups are hydrolytically removed from MCPs by the phosphorylated CheB protein, a methylesterase. These enzymes are part of a feedback circuit that stops motor responses a short time after they have commenced. The attractant-MCP complex is a good substrate for CheR and a poor substrate for CheB. When an attractant binds to the MCP, the levels of both CheY phosphate and CheB phosphate drop because the autophosphorylation of

CheA is inhibited. This not only causes counterclockwise rotation and a run, but also lowers methyltransferase activity so that MCP methylation increases. Increased methylation changes the conformation of the MCP so that it again supports an intermediate level of CheA autophosphorylation. CheY phosphate and CheB phosphate return to intermediate levels and restore the normal run-tumble behavior. Removal of the attractant causes the over-methylated MCP to stimulate CheA autophosphorylation and the levels of CheY phosphate and CheB phosphate increase. This induces tumbling and simultaneously promotes MCP demethylation so that the system returns to an intermediate level of CheA autophosphorylation.

The chemotactic response is a very complex one involving many different proteins and two forms of covalent protein regulation (*see section 8.9*). The actual response arises from a combination of (1) the control of CheA phosphorylation by attractant and repellent levels; (2) the clockwise rotation promoted by phosphorylated CheY; and (3) a feedback regulatory circuit involving CheR, phosphorylated CheB, and variations in MCP methylation.

1. What is a two-component phosphorelay system?
2. Explain in a general way how bacteria are attracted to substances like nutrients while being repelled by toxic materials.
3. Describe the molecular mechanism by which molecules attract *E. coli*.

12.7 Control of the Cell Cycle

Although much progress has been made in understanding the control of microbial enzyme activity and pathway function, much less is known about the regulation of more complex events such as bacterial sporulation and cell division. This section briefly describes the regulation of bacterial cell division. Attention is focused primarily on *E. coli* because it has been intensively studied.

The complete sequence of events extending from the formation of a new cell through the next division is called the **cell cycle**. A young *E. coli* cell growing at a constant rate will double in length without changing in diameter, then divide into two cells of equal size by transverse fission. Because each daughter cell receives at least one copy of the genetic material, DNA replication and cell division must be tightly coordinated. In fact, if DNA synthesis is inhibited by a drug or a gene mutation, cell division is also blocked and the affected cells continue to elongate, forming long filaments. Termination of DNA replication also seems connected in some way with cell division. Although the growth rate of *E. coli* at 37°C may vary considerably, division usually takes place about 20 minutes after replication has finished. During this final interval the genetic material must be distributed between the daughter cells. The newly formed DNA copies are attached to adjacent sites on the plasma membrane at or close to the center of the cell, probably at their replication factories (**figure 12.34a**). It is not yet clear how the two copies are sepa-

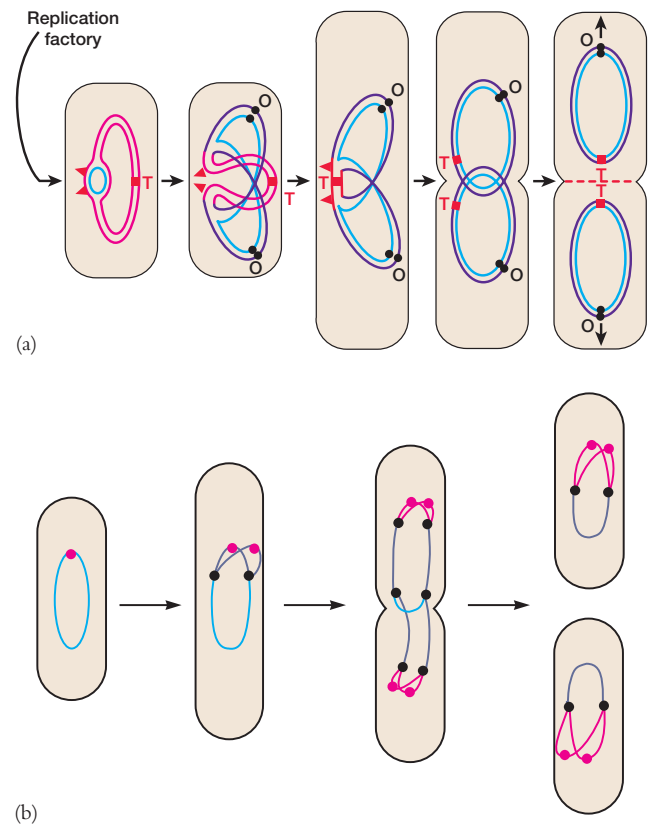


Figure 12.34 DNA Replication in Bacteria. (a) In slowly growing bacteria, the chromosome is replicated once before division. In this simplified illustration, the replicating chromosomal DNA is spooled through a membrane-bound replication factory, the factory then divides into two replication foci, and eventually the duplicated chromosomes separate and move to opposite ends of the cell. O is the origin of replication and T is the termination region. (b) DNA replication in rapidly growing bacteria is more complex. A new round of DNA replication is initiated before the original cell divides so that the DNA in daughter cells is already partially replicated. The colored circles at the ends of the DNA loops are replication origins; the black circles along the sides are replication forks. Newly synthesized DNA is in color, with the red representing the most recently synthesized DNA. Membrane attachments are not shown for sake of simplicity.

rated. ParA, ParB, MukB and other proteins are involved in DNA partitioning. The ParA and ParB proteins are localized at the poles of late predivisional cells and may be part of a mitotic-like apparatus for bacterial division. There is some evidence for active separation of chromosomes by a force-generating mechanism. Possibly the chromosomes move apart because they are pushed by the replication factories and pulled by some sort of "mitotic" apparatus. DNA movement also may result from membrane growth and cell wall synthesis, but membrane growth is too slow to account for all the movement. After the chromosomes have been separated, a cross wall or septum forms between them. [Patterns of cell wall formation \(p. 223\)](#)

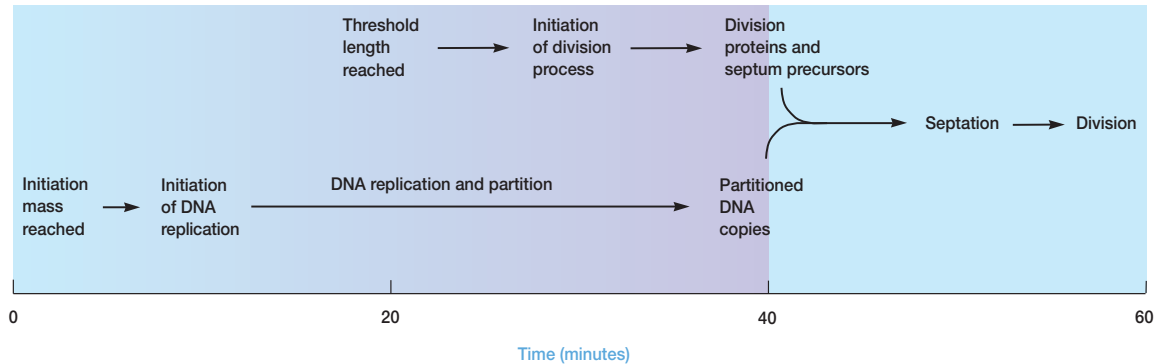


Figure 12.35 Control of the Cell Cycle in *E. coli*. A 60-minute interval between divisions has been assumed for purposes of simplicity (the actual time between cell divisions may be shorter). *E. coli* requires about 40 minutes to replicate its DNA and 20 minutes after termination of replication to prepare for division. The position of events on the time line is approximate and meant to show the general pattern of occurrences.

Current evidence suggests that two sequences of events, operating in parallel but independently, control division and the cell cycle (**figure 12.35**). Like eucaryotic cells, bacteria must reach a specific threshold size or initiation mass to trigger DNA replication. *E. coli* also has to reach a threshold length before it can partition its chromosomes and divide into two cells. Thus there seem to be two separate controls for the cell cycle, one sensitive to cell mass and the other responding to cell length. DNA replication takes about 40 minutes to complete.

Some of the *E. coli* cell cycle control mechanisms are becoming clearer, although much remains to be learned. The initiation of DNA replication requires binding of many copies of the DnaA protein to *oriC*, the replication origin site (*see pp.* 235–39). Active DnaA protein has bound ATP, and the interconversion between DnaA-ATP and DnaA-ADP may help regulate initiation. Other factors also appear to participate in initiating DNA replication. After DNA replication is under way, another round does not immediately begin, partly because the parental DNA strand is methylated right after replication. The methylated replication origin binds to specific areas on the plasma membrane and is inactive.

The initiation of septation is equally complex and tightly regulated. Both termination of DNA replication and the attainment of threshold length are required to trigger septation and cell division. This is at least partly due to the inhibition of septation by the proximity of chromosomes. The presence of DNA damage inhibits septation as well. A cell will complete chromosome replication, repair any DNA damage, and partition the chromosomes into opposite ends before it forms a septum and divides. There probably are one or more regulatory proteins that interact with various division proteins to promote septum formation and division. An adequate supply of peptidoglycan chains or precursors also must be available at the proper time. The FtsZ protein is a division protein essential in

initiating septation. This protein is scattered throughout the cell between divisions. At the onset of septation, it forms a Z ring at the septation site, and the ring then becomes smaller. Because the FtsZ protein hydrolyzes GTP, the ring is a contractile structure that uses GTP energy; it also determines the placement of the septum. Several other proteins also are required for cell division. For example, the PBP3 or penicillin-binding protein 3 catalyzes peptidoglycan transglycosylation and peptidoglycan transpeptidation to help create the new cell wall (*see pp.* 221–23). Clearly regulation of the bacterial cell cycle is complex and involves several interacting regulatory mechanisms.

The relationship of DNA synthesis to the cell cycle varies with the growth rate. If *E. coli* is growing with a doubling time of about 60 minutes, DNA replication does not take place during the last 20 minutes—that is, replication is a discontinuous process when the doubling time is 60 minutes or longer. When the culture is growing with a doubling time of less than 60 minutes, a second round of replication begins while the first round is still under way (*figure 12.34b*). The daughter cells may actually receive DNA with two or more replication forks, and replication is continuous because the cells are always copying their DNA.

Two decades of research have provided a fairly adequate overall picture of the cell cycle in *E. coli*. Several cell division genes have been identified. Yet it still is not known precisely how the cycle is controlled. Future work should improve our understanding of this important process.

1. What is a cell cycle? Briefly describe how the cycle in *E. coli* is regulated and how cycle timing results.
2. How are the two DNA copies separated and apportioned between the two daughter cells?

Summary

1. Prokaryotic mRNA has nontranslated leader and trailer sequences at its ends. Spacer regions exist between genes when mRNA is polygenic.
2. RNA is synthesized by RNA polymerase that copies the sequence of the DNA template strand (**figure 12.2**).
3. The sigma factor helps the prokaryotic RNA polymerase bind to the promoter region at the start of a gene.
4. A terminator marks the end of a gene. A rho factor is needed for RNA polymerase release from some terminators.
5. RNA polymerase II synthesizes heterogeneous nuclear RNA, which then undergoes posttranscriptional modification by RNA cleavage and addition of a 3' poly-A sequence and a 5' cap to generate eukaryotic mRNA (**figure 12.5**).
6. Many eukaryotic genes are split or interrupted genes that have exons and introns. Exons are joined by RNA splicing. Splicing involves small nuclear RNA molecules, spliceosomes, and sometimes ribozymes.
7. In translation, ribosomes attach to mRNA and synthesize a polypeptide beginning at the N-terminal end. A polysome or polyribosome is a complex of mRNA with several ribosomes.
8. Amino acids are activated for protein synthesis by attachment to the 3' end of transfer RNAs. Activation requires ATP, and the reaction is catalyzed by aminoacyl-tRNA synthetases.
9. Ribosomes are large, complex organelles composed of rRNAs and many polypeptides. Amino acids are added to a growing peptide chain at the translational domain.
10. Protein synthesis begins with the binding of fMet-tRNA (prokaryotes) or an initiator methionyl-tRNA^{Met} (eukaryotes) to an initiator codon on mRNA and to the two ribosomal subunits. This involves the participation of protein initiation factors (**figure 12.14**).
11. In the elongation cycle the proper aminoacyl-tRNA binds to the A site with the aid of EF-Tu and GTP (**figure 12.15**). Then the transpeptidation reaction is catalyzed by peptidyl transferase. Finally, during translocation, the peptidyl-tRNA moves to the P site and the ribosome travels along the mRNA one codon. Translocation requires GTP and EF-G or translocase. The empty tRNA leaves the ribosome by way of the exit site.
12. Protein synthesis stops when a nonsense codon is reached. Prokaryotes require three release factors for codon recognition and ribosome dissociation from the mRNA.
13. Molecular chaperones help proteins fold properly, protect cells against environmental stresses, and transport proteins across membranes.
14. Prokaryotic proteins may not fold until completely synthesized, whereas eukaryotic protein domains fold as they leave the ribosome. Some proteins are self-splicing and excise portions of themselves before folding into their final shape.
15. β -Galactosidase is an inducible enzyme whose concentration rises in the presence of its inducer.
16. Many biosynthetic enzymes are repressible enzymes whose levels are reduced in the presence of end products called corepressors.
17. Induction and repression result from regulation of the rate of transcription by repressor proteins coded for by regulator genes. This is an example of negative control. A regulator gene mutation can lead to a constitutive mutant, which continuously produces a metabolite.
18. The repressor inhibits transcription by binding to an operator and interfering with the binding of RNA polymerase to its promoter (**figure 12.22**).
19. In inducible systems the newly synthesized repressor protein is active, and inducer binding inactivates it. In contrast, an inactive repressor or aporepressor is synthesized in a repressible system and is activated by the corepressor (**figure 12.23**).
20. Often one repressor regulates the synthesis of several enzymes because they are part of a single operon, a DNA sequence coding for one or more polypeptides and the operator controlling its expression.
21. Positive operon control of the *lac* operon is due to the catabolite activator protein, which is activated by cAMP (**figure 12.27**).
22. In the tryptophan operon a leader region lies between the operator and the first structural gene (**figure 12.29**). It codes for the synthesis of a leader peptide and contains an attenuator, a rho-independent termination site.
23. The synthesis of the leader peptide by a ribosome while RNA polymerase is transcribing the leader region regulates transcription; therefore the tryptophan operon is expressed only when there is insufficient tryptophan available. This mechanism of transcription control is called attenuation (**figure 12.30**).
24. Global regulatory systems can control many operons simultaneously and help prokaryotes respond rapidly to a wide variety of environmental challenges.
25. Catabolite repression probably contributes to diauxic growth when *E. coli* is cultured in the presence of both glucose and lactose.
26. The transcription of genes can be regulated by altering the promoters to which RNA polymerase binds by changing the available sigma factors. A good example is the control of sporulation.
27. Small antisense RNA molecules regulate the expression of some genes. They can affect DNA replication, RNA transcription, or translation. For example, they help control porin protein levels.
28. Two-component phosphorelay systems are signal transduction systems that use phosphoryl group transfers in regulation. They have a sensor kinase and a response regulator.
29. Sporulation and chemotaxis regulatory systems are two-component phosphorelay systems.
30. The complete sequence of events extending from the formation of a new cell through the next division is called the cell cycle.
31. The end of DNA replication is tightly linked to cell division, so division in *E. coli* usually takes place about 20 minutes after replication is finished (**figure 12.34**). Special division and regulatory proteins are involved.
32. In very rapidly dividing bacterial cells, a new round of DNA replication begins before the cells divide.

Key Terms

amino acid activation 266
aminoacyl or acceptor site (A site) 270
aminoacyl-tRNA synthetases 267
anticodon triplet 266
antisense RNA 283
aporepressor 276
attenuation 281
attenuator 279

catabolite activator protein (CAP) 278
catabolite repression 281
cell cycle 285
constitutive mutant 276
core enzyme 262
corepressor 276
3', 5'-cyclic adenosine monophosphate (cAMP) 278

cyclic AMP receptor protein (CRP) 278
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rho factor 263

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sigma factor 262
small nuclear RNA (snRNA) 264
spliceosome 264
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structural gene 277
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transfer RNA (tRNA) 261
translocation 270
transpeptidation reaction 270
two-component phosphorelay system 283

Questions for Thought and Review

1. Describe how RNA polymerase transcribes procaryotic DNA. How does the polymerase know where to begin and end transcription?
2. How do eucaryotic RNA polymerases and promoters differ from those in procaryotes? In what ways does eucaryotic mRNA differ from procaryotic mRNA with respect to synthesis and structure? How does eucaryotic synthesis of rRNA and tRNA resemble that of mRNA? How does it differ?
3. Draw diagrams summarizing the sequence of events in the three stages of protein synthesis (initiation, elongation, and termination) and accounting for the energy requirements of translation.
4. Describe in some detail the organization of the regulatory systems responsible for induction and repression, and the mechanism of their operation.
5. How is *E. coli* able to use glucose exclusively when presented with a mixture of glucose and lactose?
6. Of what practical importance is attenuation in coordinating the synthesis of amino acids and proteins? Describe how attenuation activity would vary when protein synthesis suddenly rapidly accelerated, then later suddenly decelerated.
7. How does the timing of DNA replication seem to differ between slow-growing and fast-growing cells? Be able to account for the fact that bacterial cells may contain more than a single copy of DNA.

Critical Thinking Questions

1. Attenuation affects anabolic pathways, whereas repression can affect either anabolic or catabolic pathways. Provide an explanation for this.
2. Many people say that RNA was the first of the information molecules (RNA, DNA, protein) to occur during evolution. Given the information in this chapter, what evidence is there to support this hypothesis?
3. Compare and contrast RNA and DNA synthesis.

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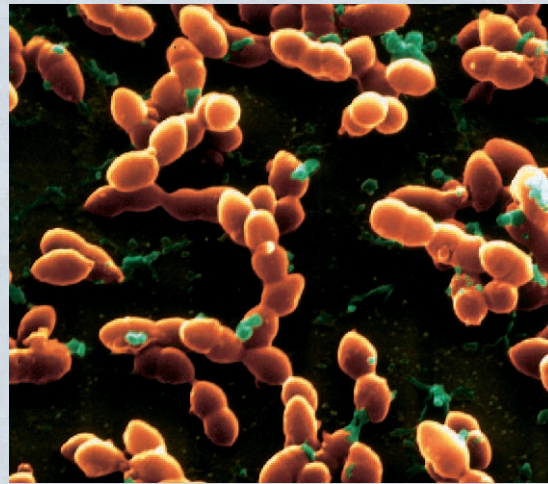
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CHAPTER 13

Microbial Recombination and Plasmids



The scanning electron micrograph shows *Streptococcus pneumoniae*, the bacterium first used to study transformation and obtain evidence that DNA is the genetic material of organisms.

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Concepts

1. Recombination is a one-way process in prokaryotes: a piece of genetic material (the exogenote) is donated to the chromosome of a recipient cell (the endogenote) and integrated into it.
2. The actual transfer of genetic material between bacteria usually takes place in one of three ways: direct transfer between two bacteria temporarily in physical contact (conjugation), transfer of a naked DNA fragment (transformation), or transport of bacterial DNA by bacteriophages (transduction).
3. Plasmids and transposable elements can move genetic material between bacterial chromosomes and within chromosomes to cause rapid changes in genomes and drastically alter phenotypes.
4. The bacterial chromosome can be mapped with great precision, using Hfr conjugation in combination with transformational and transductional mapping techniques.
5. Recombination of virus genomes occurs when two viruses with homologous chromosomes infect a host cell at the same time.

*Deep in the cavern of the infant's breast
The father's nature lurks, and lives anew.*

—Horace, Odes

Chapter 12 introduces the fundamentals of molecular genetics: the way genetic information is organized and stored, the nature of mutations and techniques for their isolation and study, and DNA repair. This chapter focuses on genetic recombination in microorganisms, with primary emphasis placed on recombination in bacteria and viruses.

The chapter begins with a general overview of bacterial recombination and an introduction to both bacterial plasmids and transposable elements. Next, the three types of bacterial gene transfer—conjugation, transformation, and transduction—are discussed. Because an understanding of the techniques used to locate genes on chromosomes depends on knowledge of recombination mechanisms, mapping the bacterial genome is discussed after the introduction to recombination. The chapter ends with a description of recombination in viruses and a brief discussion of viral chromosome mapping.

In a general sense, **recombination** is the process in which one or more nucleic acids molecules are rearranged or combined to produce a new nucleotide sequence. Usually genetic material from two parents is combined to produce a recombinant chromosome with a new, different genotype. Recombination results in a new arrangement of genes or parts of genes and normally is accompanied by a phenotypic change. Most eucaryotes exhibit a complete sexual life cycle, including meiosis, a process of extreme importance in generating new combinations of alleles (alternate forms of a particular gene) through recombination. These chromosome exchanges during meiosis result from **crossing-over** between homologous chromosomes, chromosomes containing identical sequences of genes (**figure 13.1**). Until about 1945 the primary focus in genetic analysis was on the recombination of genes in plants and animals. The early work on recombination in higher eucaryotes laid the foundations of classical genetics, but it was the development of bacterial and phage genetics between about 1945 and 1965 that really stimulated a rapid advance in our understanding of molecular genetics. [Meiosis \(pp. 87–88\)](#)

13.1 Bacterial Recombination: General Principles

Microorganisms carry out several types of recombination. **General recombination**, the most common form, usually involves a reciprocal exchange between a pair of homologous DNA sequences. It can occur anywhere on the chromosome, and it results from DNA strand breakage and reunion leading to crossing-over (**figure 13.2**). General recombination is carried out by the prod-

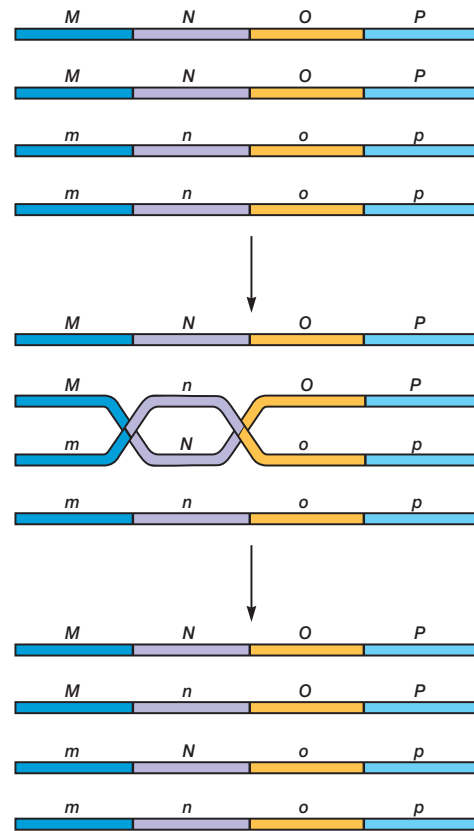


Figure 13.1 Crossing-Over. An example of recombination through crossing-over between homologous eucaryotic chromosomes. The *Nn* gene pair is exchanged. This process usually occurs during meiosis.

ucts of *rec* genes such as the *recA* protein so important for DNA repair (*see pp. 254–55*). In bacterial transformation a nonreciprocal form of general recombination takes place (**figure 13.3**). A piece of genetic material is inserted into the chromosome through the incorporation of a single strand to form a stretch of **heteroduplex DNA**. A second type of recombination, one particularly important in the integration of virus genomes into bacterial chromosomes, is **site-specific recombination**. The genetic material is not homologous with the chromosome it joins, and generally the enzymes responsible for this event are specific for the particular virus and its host. A third kind of recombination, which may be considered a type of site-specific recombination, is called **replicative recombination**. It accompanies the replication of genetic material and does not depend on sequence homology. It is used by some genetic elements that move about the chromosome. [DNA replication \(pp. 235–39\)](#)

Although sexual reproduction with the formation of a zygote and subsequent meiosis is not present in bacteria, recombination can take place in several ways following **horizontal gene transfer**. In this process genes are transferred from one independent, mature organism to another. Horizontal gene transfer is quite different

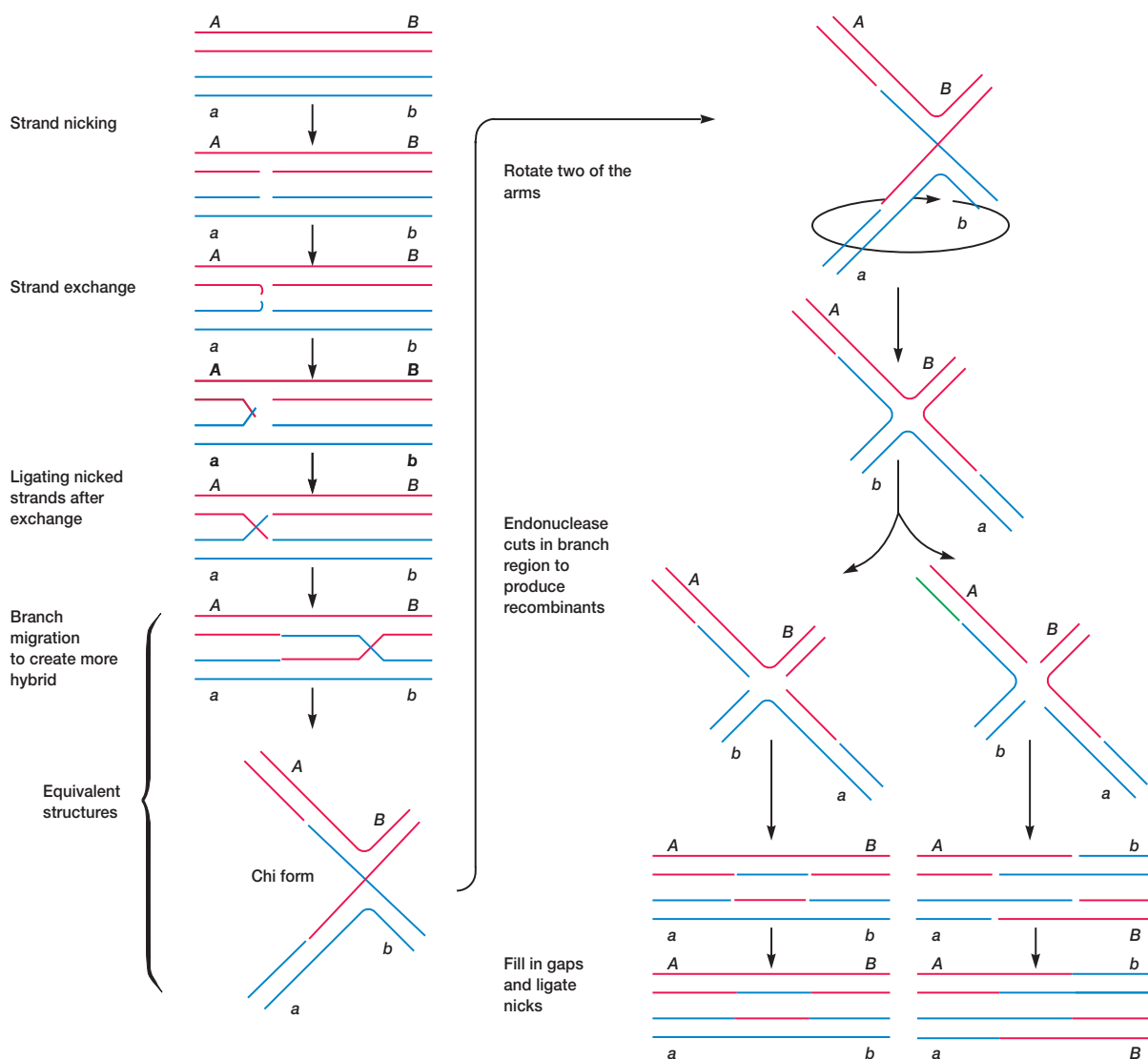


Figure 13.2 The Holliday Model for Reciprocal General Recombination. Source: From H. Potter and D. Dressler; Proceedings of the National Academy of Sciences, 73:3000, 1976; after R. Holliday, Genetics, 78:273, 1974; and previous publications cited therein.

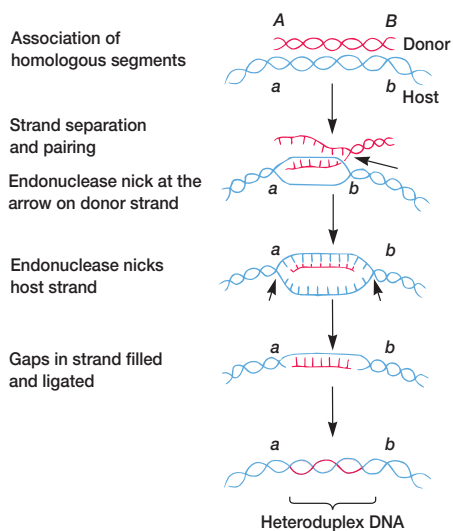


Figure 13.3 Nonreciprocal General Recombination. The Fox model for nonreciprocal general recombination. This mechanism has been proposed for the recombination occurring during transformation in some bacteria.

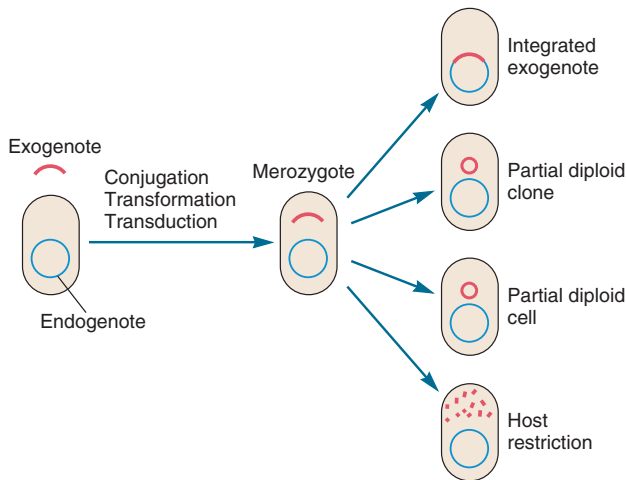


Figure 13.4 The Production and Fate of Merozygotes. See text for discussion.

from the transmission of genes from parents to offspring (vertical gene transfer). In general, a piece of donor DNA, the **exogenote**, must enter the recipient cell and become a stable part of the recipient cell's genome, the **endogenote**. Two kinds of DNA can move between bacteria. If a DNA fragment is the exchange vehicle, then the exogenote must get into the recipient cell and become incorporated into the endogenote as a replacement piece (or as an "extra" piece) without being destroyed by the host. During replacement of host genetic material, the recipient cell becomes temporarily diploid for a portion of the genome and is called a **merozygote** (figure 13.4). Sometimes the DNA exists in a form that cannot be degraded by the recipient cell's endonucleases. In this case the DNA does not need to be integrated into the host genome but must only enter the recipient to confer its genetic information on the cell. Most linear DNA fragments are not stably maintained unless they have been integrated into the bacterial genome. Resistant DNA, such as that in plasmids (see following), usually is circular and has sequences that allow it to maintain itself independent of the host chromosome. Recombination in bacteria is a one-way gene transfer from donor to recipient. Recombination in eucaryotes is reciprocal—that is, all of the DNA is conserved in the gametes that eventually arise from meiosis and recombination.

Movement of DNA from a donor bacterium to the recipient can take place in three ways: direct transfer between two bacteria temporarily in physical contact (conjugation), transfer of a naked DNA fragment (transformation), and transport of bacterial DNA by bacteriophages (transduction). Whatever the mode of transfer, the exogenote has only four possible fates in the recipient (figure 13.4). First, when the exogenote has a sequence homologous to that of the endogenote, integration may occur; that is, it may pair with the recipient DNA and be incorporated to yield a recombinant genome. Second, the foreign DNA some-

times persists outside the endogenote and replicates to produce a clone of partially diploid cells. Third, the exogenote may survive, but not replicate, so that only one cell is a partial diploid. Finally, host cell nucleases may degrade the exogenote, a process called **host restriction**.

1. Define the following terms: recombination, crossing-over, general recombination, site-specific recombination, replicative recombination, exogenote, endogenote, horizontal gene transfer, merozygote, and host restriction.
2. Distinguish among the three forms of recombination mentioned in this section.
3. What four fates can DNA have after entering a bacterium?

13.2 Bacterial Plasmids

Conjugation, the transfer of DNA between bacteria involving direct contact, depends on the presence of an "extra" piece of circular DNA known as a plasmid. Plasmids play many important roles in the lives of bacteria. They also have proved invaluable to microbiologists and molecular geneticists in constructing and transferring new genetic combinations and in cloning genes (see chapter 14). In this section the different types of bacterial plasmids are discussed.

Plasmids are small double-stranded DNA molecules, usually circular, that can exist independently of host chromosomes and are present in many bacteria (they are also present in some yeasts and other fungi). They have their own replication origins and are autonomously replicating and stably inherited. A **replicon** is a DNA molecule or sequence that has a replication origin and is capable of being replicated. Plasmids and bacterial chromosomes are separate replicons. Plasmids have relatively few genes, generally less than 30. Their genetic information is not essential to the host, and bacteria that lack them usually function normally. Single-copy plasmids produce only one copy per host cell. Multicopy plasmids may be present at concentrations of 40 or more per cell.

Characteristically, plasmids can be eliminated from host cells in a process known as **curing**. Curing may occur spontaneously or be induced by treatments that inhibit plasmid replication while not affecting host cell reproduction. The inhibited plasmids are slowly diluted out of the growing bacterial population. Some commonly used curing treatments are acridine mutagens, UV and ionizing radiation, thymine starvation, and growth above optimal temperatures.

Plasmids may be classified in terms of their mode of existence and spread. An **episome** is a plasmid that can exist either with or without being integrated into the host's chromosome. Some plasmids, **conjugative plasmids**, have genes for pili and can transfer copies of themselves to other bacteria during conjugation. A brief summary of the types of plasmids and their properties is given in table 13.1.

Table 13.1 Major Types of Plasmids

Type	Representatives	Approximate Size (kbp)	Copy Number (Copies/Chromosome)	Hosts	Phenotypic Features ^a
Fertility Factor^b	F factor	95–100	1–3	<i>E. coli</i> , <i>Salmonella</i> , <i>Citrobacter</i>	Sex pilus, conjugation
R Plasmids	RP4	54	1–3	<i>Pseudomonas</i> and many other gram-negative bacteria	Sex pilus, conjugation, resistance to Ap, Km, Nm, Tc
	R1	80	1–3	Gram-negative bacteria	Resistance to Ap, Km, Su, Cm, Sm
	R6	98	1–3	<i>E. coli</i> , <i>Proteus mirabilis</i>	Su, Sm, Cm, Tc, Km, Nm
	R100	90	1–3	<i>E. coli</i> , <i>Shigella</i> , <i>Salmonella</i> , <i>Proteus</i>	Cm, Sm, Su, Tc, Hg
	pSH6	21		<i>Staphylococcus aureus</i>	Gm, Tm, Km
	pSJ23a	36		<i>S. aureus</i>	Pn, Asa, Hg, Gm, Km, Nm, Em, etc.
	pAD2	25		<i>Enterococcus faecalis</i>	Em, Km, Sm
Col Plasmids	ColE1	9	10–30	<i>E. coli</i>	Colicin E1 production
	ColE2		10–15	<i>Shigella</i>	Colicin E2
	CloDF13			<i>Enterobacter cloacae</i>	Cloacin DF13
Virulence Plasmids	Ent (P307)	83		<i>E. coli</i>	Enterotoxin production
	K88 plasmid			<i>E. coli</i>	Adherence antigens
	ColV-K30	2		<i>E. coli</i>	Siderophore for iron uptake; resistance to immune mechanisms
	pZA10	56		<i>S. aureus</i>	Enterotoxin B
	Ti	200		<i>Agrobacter tumefaciens</i>	Tumor induction
Metabolic Plasmids	CAM	230		<i>Pseudomonas</i>	Camphor degradation
	SAL	56		<i>Pseudomonas</i>	Salicylate degradation
	TOL	75		<i>Pseudomonas putida</i>	Toluene degradation
	pJP4			<i>Pseudomonas</i>	2,4-dichlorophenoxyacetic acid degradation
				<i>E. coli</i> , <i>Klebsiella</i> , <i>Salmonella</i>	Lactose degradation
				<i>Providencia</i>	Urease
				<i>Rhizobium</i>	Nitrogen fixation and symbiosis
	sym				

^aAbbreviations used for resistance to antibiotics and metals: Ap, ampicillin; Asa, arsenate; Cm, chloramphenicol; Em, erythromycin; Gm, gentamycin; Hg, mercury; Km, kanamycin; Nm, neomycin; Pn, penicillin; Sm, streptomycin; Su, sulfonamides; Tc, tetracycline.

^bMany R plasmids, metabolic plasmids, and others are also conjugative.

Fertility Factors

A plasmid called the fertility or **F factor** plays a major role in conjugation in *E. coli* and was the first to be described (**figure 13.5**). The F factor is about 100 kilobases long and bears genes responsible for cell attachment and plasmid transfer between specific bacterial strains during conjugation. Most of the information required for plasmid transfer is located in the *tra* operon, which contains at least 28 genes. Many of these direct the formation of

sex pili that attach the F⁺ cell (the donor cell containing an F plasmid) to an F[−] cell (**figure 13.6**). Other gene products aid DNA transfer. [Sex pili \(p. 63\)](#)

The F factor also has several segments called insertion sequences (p. 298) that assist plasmid integration into the host cell chromosome. Thus the F factor is an episome that can exist outside the bacterial chromosome or be integrated into it (**figure 13.7**).

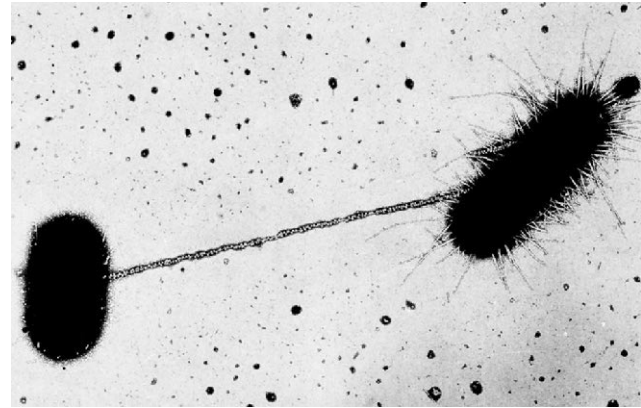
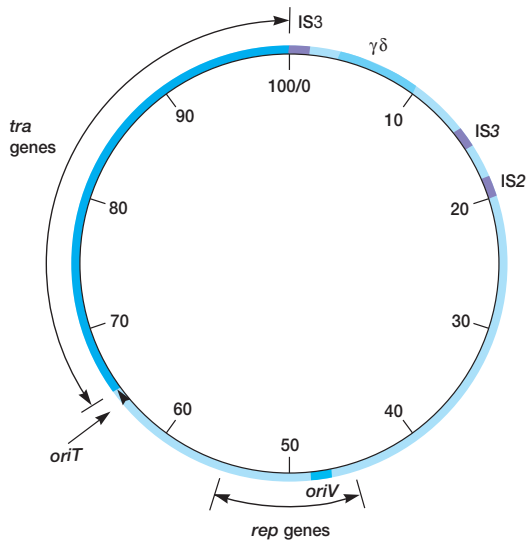


Figure 13.6 Bacterial Conjugation. An electron micrograph of two *E. coli* cells in an early stage of conjugation. The F^+ cell to the right is covered with small pili or fimbriae, and a sex pilus connects the two cells.

Figure 13.5 The F plasmid. A map showing the size and general organization of the F plasmid. The plasmid contains several transposable elements. IS2 and IS3 are insertion sequences; $\gamma\delta$ is also called transposon Tn1000. The *tra* genes code for proteins needed in pilus synthesis and conjugation. The *rep* genes code for proteins involved in DNA replication. *OriV* is the initiation site for circular DNA replication and *oriT*, the site for initiation of rolling circle replication and gene transfer during conjugation.

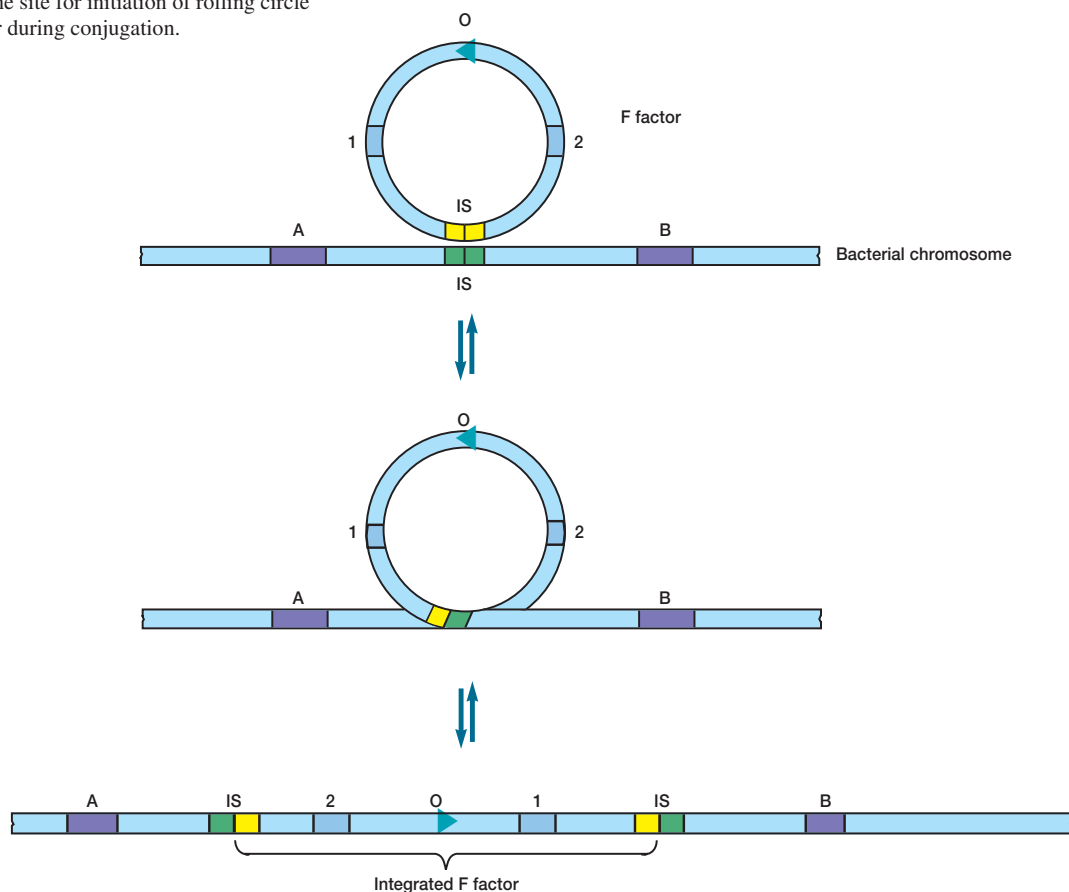


Figure 13.7 F Plasmid Integration. The reversible integration of an F plasmid or factor into a host bacterial chromosome. The process begins with association between plasmid and bacterial insertion sequences. The O arrowhead (blue-green) indicates the site at which the oriented transfer of chromosome to the recipient cell begins. A, B, 1, and 2 represent genetic markers.

Box 13.1

Virulence Plasmids and Disease

It is becoming increasingly evident that many bacteria are pathogenic because of their plasmids. These plasmids can carry genes for toxins, render the bacterium better able to establish itself in the host, or aid in resistance to host defenses. *E. coli* provides the best-studied example of virulence plasmids. Several strains of *E. coli* cause diarrhea. The enterotoxigenic strains responsible for traveler's diarrhea can produce two toxins: a heat-labile toxin (LT), which is a large protein very similar in structure and mechanism of action to cholera toxin (see chapter 32), and a heat-stable toxin (ST), a low molecular weight polypeptide. Both toxin genes are plasmid borne, and sometimes they are even carried by the same plasmid. The ST toxin gene is located on a transposon. Enterotoxigenic strains of *E. coli* also must be able to colonize the epithelium of the small intestine to cause diarrhea. This is made possible by the presence of special adhesive fimbriae encoded by genes on another plasmid. A second type of pathogenic *E. coli* invades the intestinal epithelium and causes a form of diarrhea very similar to the dysentery resulting from a *Shigella* infection. This *E. coli* strain and

Shigella contain virulence plasmids that code for special cell wall antigens and other factors enabling them to enter and destroy epithelial cells.

Some *E. coli* strains can invade the blood and organs of a host, causing a generalized infection. These pathogens often have ColV plasmids and produce colicin V. The ColV plasmid carries genes for two virulence determinants. One product increases bacterial resistance to host defense mechanisms involving complement (see sections 31.7 and 32.3). The other plasmid gene directs the synthesis of a hydroxamate that enables *E. coli* to accumulate iron more efficiently from its surroundings (see section 5.6). Since iron is not readily available in the animal host, but is essential for bacterial growth, this is an important factor in pathogenicity.

Several other pathogens carry virulence plasmids. Some *Staphylococcus aureus* strains produce an exfoliative toxin that is plasmid borne. The toxin causes the skin to loosen and often peel off in sheets, leading to the disease staphylococcal scalded skin syndrome (see section 39.3). Other plasmid-borne toxins are the tetanus toxin of *Clostridium tetani* and the anthrax toxin of *Bacillus anthracis*.

Resistance Factors

Plasmids often confer antibiotic resistance on the bacteria that contain them. **R factors** or plasmids typically have genes that code for enzymes capable of destroying or modifying antibiotics. They are not usually integrated into the host chromosome. Genes coding for resistance to antibiotics such as ampicillin, chloramphenicol, and kanamycin have been found in plasmids. Some R plasmids have only a single resistance gene, whereas others can have as many as eight. Often the resistance genes are within a transposon (p. 298), and thus it is possible for bacterial strains to rapidly develop multiple resistance plasmids. [R factors and antibiotic resistance \(p. 819\)](#)

Because many R factors also are conjugative plasmids, they can spread throughout a population, although not as rapidly as the F factor. Often, nonconjugative R factors also move between bacteria during plasmid promoted conjugation. Thus a whole population can become resistant to antibiotics. The fact that some of these plasmids are readily transferred between species further promotes the spread of resistance. When the host consumes large quantities of antibiotics, *E. coli* and other bacteria with R factors are selected for and become more prevalent. The R factors can then be transferred to more pathogenic genera such as *Salmonella* or *Shigella*, causing even greater public health problems (see section 35.7).

Col Plasmids

Bacteria also harbor plasmids with genes that may give them a competitive advantage in the microbial world. **Bacteriocins** are bacterial proteins that destroy other bacteria. They usually act only against closely related strains. Bacteriocins often kill cells

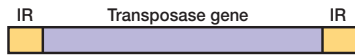
by forming channels in the plasma membrane, thus increasing its permeability. They also may degrade DNA and RNA or attack peptidoglycan and weaken the cell wall. Col plasmids contain genes for the synthesis of bacteriocins known as colicins, which are directed against *E. coli*. Similar plasmids carry genes for bacteriocins against other species. For example, Col plasmids produce cloacins that kill *Enterobacter* species. Clearly the host is unaffected by the bacteriocin it produces. Some Col plasmids are conjugative and also can carry resistance genes. [Bacteriocins and host defenses \(p. 712\)](#)

Other Types of Plasmids

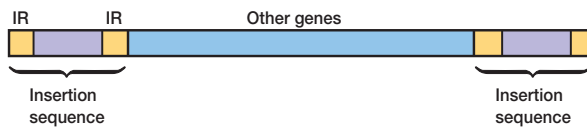
Several other important types of plasmids have been discovered. Some plasmids, called **virulence plasmids**, make their hosts more pathogenic because the bacterium is better able to resist host defense or to produce toxins. For example, enterotoxigenic strains of *E. coli* cause traveler's diarrhea because of a plasmid that codes for an enterotoxin (**Box 13.1**). **Metabolic plasmids** carry genes for enzymes that degrade substances such as aromatic compounds (toluene), pesticides (2,4-dichlorophenoxyacetic acid), and sugars (lactose). Metabolic plasmids even carry the genes required for some strains of *Rhizobium* to induce legume nodulation and carry out nitrogen fixation.

1. Give the major distinguishing features of a plasmid. What is an episome? A conjugative plasmid?
2. Describe each of the following plasmids and their importance: F factor, R factor, Col plasmid, virulence plasmid, and metabolic plasmid.

(a) Insertion sequence



(b) A composite transposon



(c) A target site for the Tn3 transposon

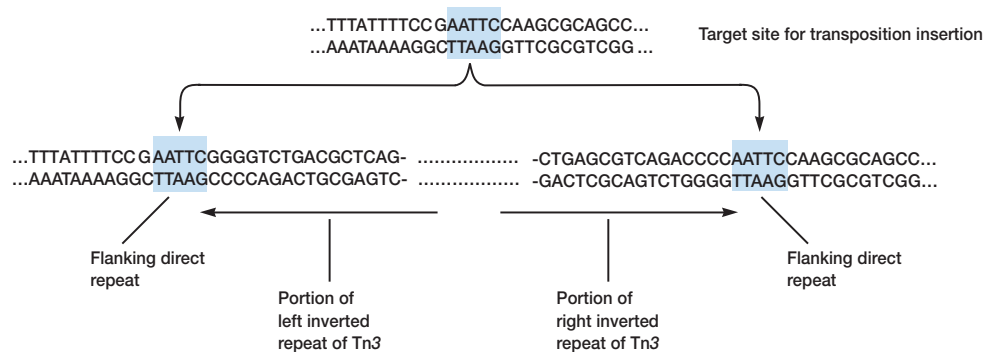


Figure 13.8 Insertion Sequences and Transposons. The structure of insertion sequences (a), composite transposons (b), and target sites (c). IR stands for inverted repeat. In (c), the highlighted five-base target site is duplicated during Tn3 transposition to form flanking direct repeats. The remainder of Tn3 lies between the inverted repeats.

13.3 Transposable Elements

The chromosomes of bacteria, viruses, and eucaryotic cells contain pieces of DNA that move around the genome. Such movement is called **transposition**. DNA segments that carry the genes required for this process and consequently move about chromosomes are **transposable elements** or **transposons**. Unlike other processes that reorganize DNA, transposition does not require extensive areas of homology between the transposon and its destination site. Transposons behave somewhat like lysogenic prophages (see pp. 390–95) except that they originate in one chromosomal location and can move to a different location in the same chromosome. Transposable elements differ from phages in lacking a virus life cycle and from plasmids in being unable to reproduce autonomously and to exist apart from the chromosome. They were first discovered in the 1940s by Barbara McClintock during her studies on maize genetics (a discovery that won her the Nobel prize in 1983). They have been most intensely studied in bacteria.

The simplest transposable elements are **insertion sequences** or **IS elements** (figure 13.8a). An IS element is a short sequence of DNA (around 750 to 1,600 base pairs [bp] in length) containing only the genes for those enzymes required for its transposi-

tion and bounded at both ends by identical or very similar sequences of nucleotides in reversed orientation known as inverted repeats (figure 13.8c). Inverted repeats are usually about 15 to 25 base pairs long and vary among IS elements so that each type of IS has its own characteristic inverted repeats. Between the inverted repeats is a gene that codes for an enzyme called **transposase** (and sometimes a gene for another essential protein). This enzyme is required for transposition and accurately recognizes the ends of the IS. Each type of element is named by giving it the prefix IS followed by a number. In *E. coli* several copies of different IS elements have been observed; some of their properties are given in table 13.2.

Transposable elements also can contain genes other than those required for transposition (for example, antibiotic resistance or toxin genes). These elements often are called **composite transposons** or **elements**. Complete agreement about the nomenclature of transposable elements has not yet been reached. Sometimes transposable elements are called transposons when they have extra genes, and insertion sequences when they lack these. Composite transposons often consist of a central region containing the extra genes, flanked on both sides by IS elements that are identical or very similar in sequence (figure 13.8b). Many com-

Table 13.2 The Properties of Selected Insertion Sequences

Insertion Sequence	Length (bp)	Inverted Repeat (Length in bp)	Target Site (Length in bp)	Number of Copies on <i>E. coli</i> Chromosome
IS1	768	23	9 or 8	6–10
IS2	1,327	41	5	4–13(1) ^a
IS3	1,400	38	3–4	5–6(2)
IS4	1,428	18	11 or 12	1–2
IS5	1,195	16	4	10–11

^aThe value in parentheses indicates the number of IS elements on the F factor.

Table 13.3 The Properties of Selected Composite Transposons

Transposon	Length (bp)	Terminal Repeat Length	Terminal Module	Genetic Markers ^a
Tn3	4,957	38		Ap
Tn501	8,200	38		Hg
Tn951	16,500	Unknown		Lactose utilization
Tn5	5,700		IS50	Km
Tn9	2,500		IS1	Cm
Tn10	9,300		IS10	Tc
Tn903	3,100		IS903	Km
Tn1681	2,061		IS1	Heat-stable enterotoxin
Tn2901	11,000		IS1	Arginine biosynthesis

^aAbbreviations for antibiotics and metals same as in table 13.1.

posite transposons are simpler in organization. They are bounded by short inverted repeats, and the coding region contains both transposition genes and the extra genes. It is believed that composite transposons are formed when two IS elements associate with a central segment containing one or more genes. This association could arise if an IS element replicates and moves only a gene or two down the chromosome. Composite transposon names begin with the prefix Tn. Some properties of selected composites are given in **table 13.3**.

The process of transposition in procaryotes involves a series of events, including self-replication and recombinational processes. Typically in bacteria, the original transposon remains at the parental site on the chromosome, while a replicated copy inserts at the target DNA (figure 13.8c). This is called replicative transposition. Target sites are specific sequences about five to nine base pairs long. When a transposon inserts at a target site, the target sequence is duplicated so that short, direct-sequence repeats flank the transposon's terminal inverted repeats (**figure 13.9**). This can be seen in figure 13.8c where the five base pair target sequence moves to both ends of the transposon and retains the same orientation.

The transposition of the Tn3 transposon is a well-studied example of replicative transposition. Its mechanism is outlined in **figure 13.10**. In the first stage the plasmid containing the Tn3 transposon fuses with the target plasmid to form a cointegrate molecule (figure 13.10, steps 1 to 4). This process requires the Tn3 trans-

posase enzyme coded for by the *tnpA* gene (**figure 13.11**). Note that the cointegrate has two copies of the Tn3 transposon. In the second stage the cointegrate is resolved to yield two plasmids, each with a copy of the transposon (figure 13.10, steps 5 and 6). Resolution involves a crossover at the two *res* sites and is catalyzed by a resolvase enzyme coded for by the *tnpR* gene (figure 13.11).

Transposable elements produce a variety of important effects. They can insert within a gene to cause a mutation or stimulate DNA rearrangement, leading to deletions of genetic material. If a transposon insertion produces an obvious phenotypic change, the gene can be tracked by following this altered phenotype. One can fragment the genome and isolate the mutated fragment, thereby partially purifying the gene. Thus transposons may be used to purify genes and study their functions. Because some transposons carry stop codons or termination sequences, they may block translation or transcription. Other elements carry promoters and thus activate genes near the point of insertion. Eucaryotic genes as well as procaryotic genes can be turned on and off by transposon movement. Transposons also are located within plasmids and participate in such processes as plasmid fusion and the insertion of F plasmids into the *E. coli* chromosome (figure 13.7).

In the previous discussion of plasmids, it was noted that an R plasmid can carry genes for resistance to several drugs. Transposons have antibiotic resistance genes and play a major role in generating these plasmids. Consequently the existence of these elements causes

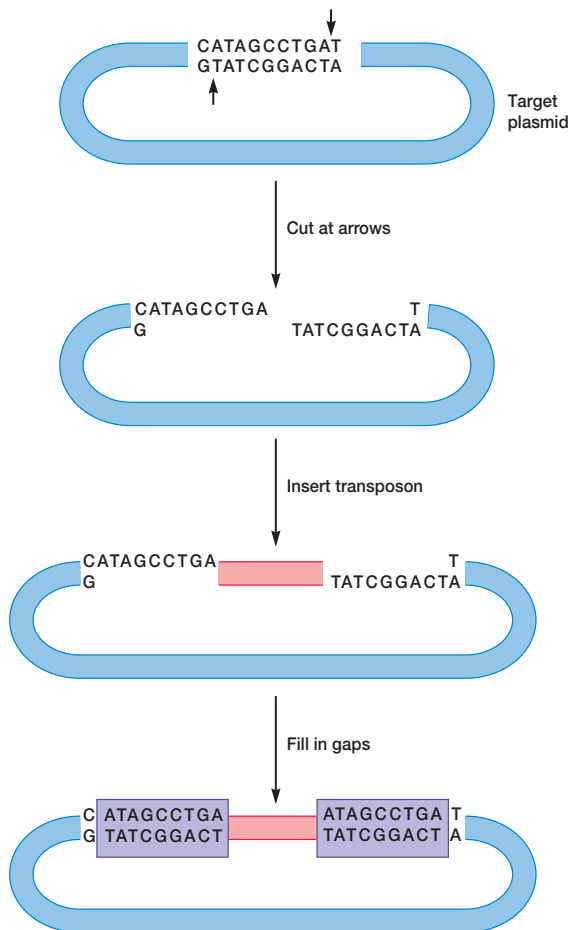


Figure 13.9 Generation of Direct Repeats in Host DNA Flanking a Transposon. (a) The arrows indicate where the two strands of host DNA will be cut in a staggered fashion, 9 base pairs apart. (b) After cutting. (c) The transposon (pink) has been ligated to one strand of host DNA at each end, leaving two 9 base gaps. (d) After the gaps are filled in, there are 9 base pair repeats of host DNA (purple boxes) at each end of the transposon.

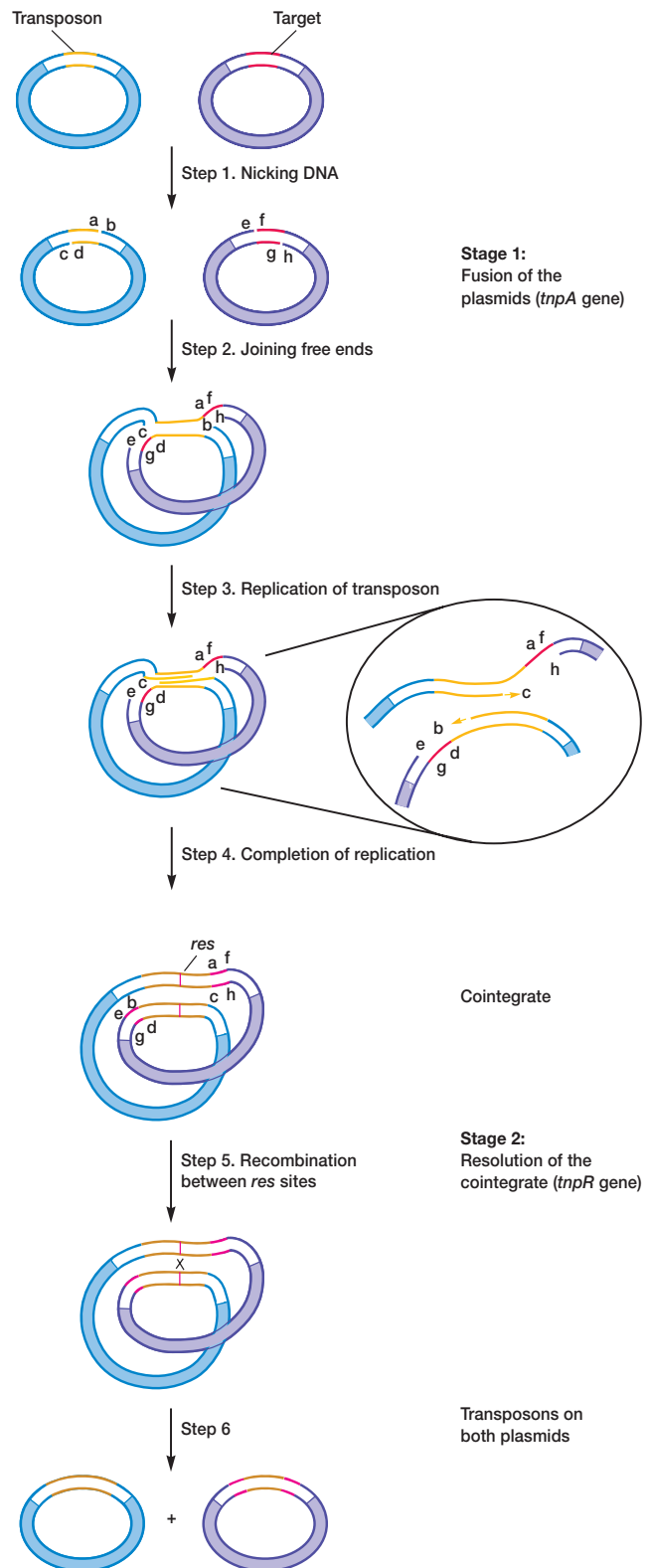


Figure 13.10 Tn3 Transposition Mechanism. Step 1: The two plasmids are nicked to form the free ends labeled a–h. Step 2: Ends a and f are joined, as are g and d. This leaves b, c, e, and h free. Step 3: Two of these remaining free ends (b and c) serve as primers for DNA replication, which is shown in a blowup of the replicating region. Step 4: Replication continues until end b reaches e and end c reaches h. These ends are ligated to complete the cointegrate. Notice that the whole transposon has been replicated. The paired *res* sites are shown for the first time here, even though one *res* site existed in the previous steps. Steps 5 and 6: A crossover occurs between the two *res* sites in the two copies of the transposon, leaving two independent plasmids, each bearing a copy of the transposon.

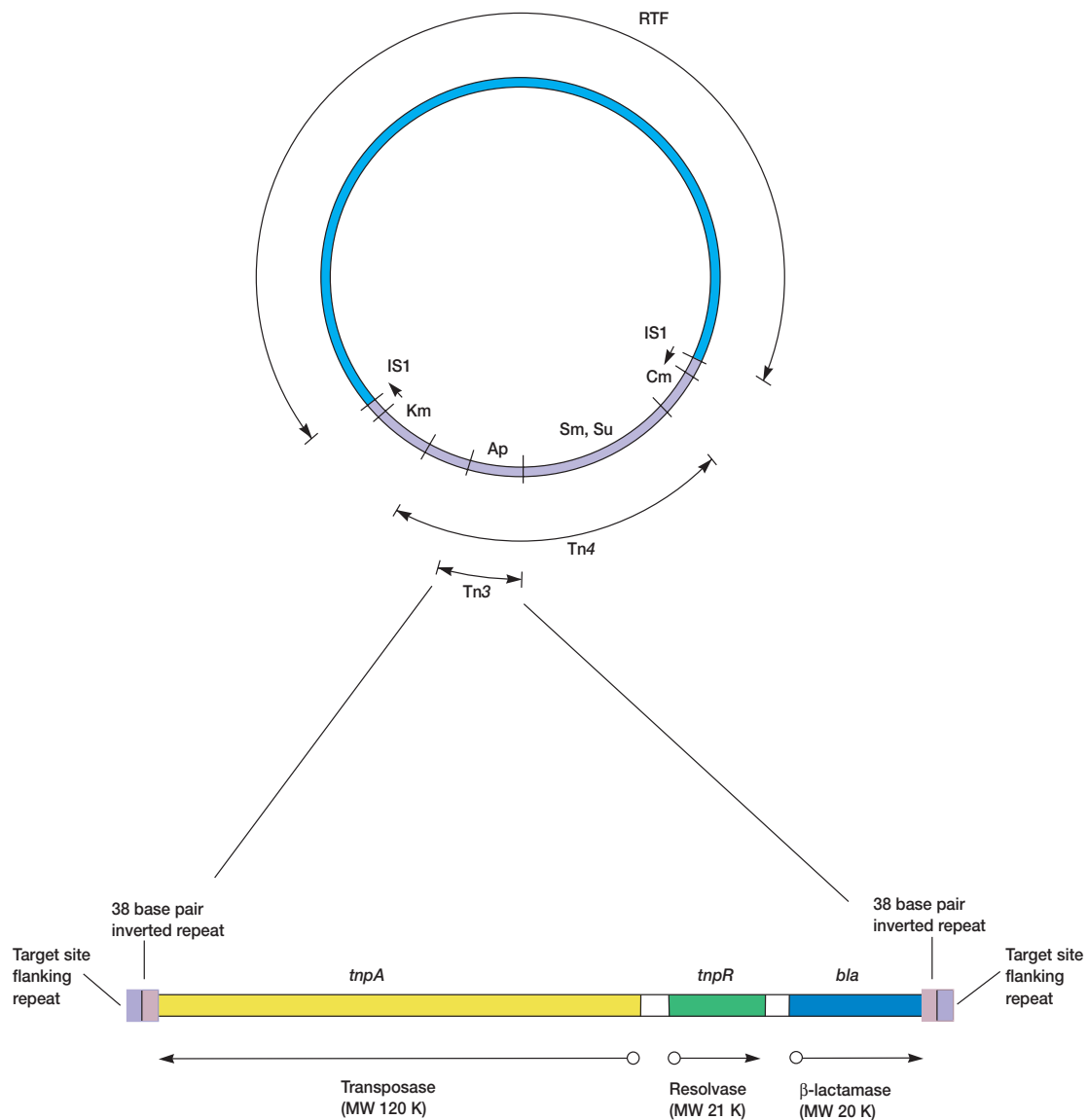


Figure 13.11 The Structure of R Plasmids and Transposons. The R1 plasmid carries resistance genes for five antibiotics: chloramphenicol (Cm), streptomycin (Sm), sulfonamide (Su), ampicillin (Ap), and kanamycin (Km). These are contained in the Tn3 and Tn4 transposons. The resistance transfer factor (RTF) codes for the proteins necessary for plasmid replication and transfer. The structure of Tn3 is shown in more detail. The arrows indicate the direction of gene transcription.

serious problems in the treatment of disease. Since plasmids can contain several different target sites, transposons will move between them; thus plasmids act as both the source and the target for transposons with resistance genes. In fact, multiple drug resistance plasmids probably often arise from transposon accumulation on a single plasmid (figure 13.11). Because transposons also move between plasmids and primary chromosomes, drug resistance genes can exchange between plasmids and chromosomes, resulting in the further spread of antibiotic resistance.

Some transposons bear transfer genes and can move between bacteria through the process of conjugation, as discussed in the next section. A well-studied example of such a **conjugative transposon** is Tn916 from *Enterococcus faecalis*. Although Tn916 cannot replicate autonomously, it will transfer itself from *E. faecalis* to a variety of recipients and integrate into their chromosomes. Because it carries a gene for tetracycline resistance, this conjugative transposon also spreads drug resistance.

Transposable elements are widespread in nature. They are present in eucaryotes, bacteria, and the Archaea. For example, transposable elements have been found in yeast, maize, *Drosophila*, and humans. Clearly, transposable elements play an extremely important role in the generation and transfer of new gene combinations.

1. Define the following terms: transposition, transposable element or transposon, insertion sequence, transposase, conjugative transposon, and composite transposon. Be able to distinguish between an insertion sequence and a composite transposon.
2. How does transposition usually occur in bacteria, and what happens to the target site? What is replicative transposition?
3. Give several important effects transposable elements have on bacteria.

13.4 Bacterial Conjugation

The initial evidence for bacterial **conjugation**, the transfer of genetic information by direct cell to cell contact, came from an elegant experiment performed by Joshua Lederberg and Edward L. Tatum in 1946. They mixed two auxotrophic strains, incubated the culture for several hours in nutrient medium, and then plated it on minimal medium. To reduce the chance that their results were due to simple reversion, they used double and triple auxotrophs on the assumption that two or three reversions would not often occur simultaneously. For example, one strain required biotin (Bio⁻), phenylalanine (Phe⁻), and cysteine (Cys⁻) for growth, and another needed threonine (Thr⁻), leucine (Leu⁻), and thiamine (Thi⁻). Recombinant prototrophic colonies appeared on the minimal medium after incubation (**figure 13.12**). Thus the chromosomes of the two auxotrophs were able to associate and undergo recombination.

Lederberg and Tatum did not directly prove that physical contact of the cells was necessary for gene transfer. This evidence was provided by Bernard Davis (1950), who constructed a U tube consisting of two pieces of curved glass tubing fused at the base to form a U shape with a fritted glass filter between the halves. The filter allows the passage of media but not bacteria. The U tube was filled with nutrient medium and each side inoculated with a different auxotrophic strain of *E. coli* (**figure 13.13**). During incubation, the medium was pumped back and forth through the filter to ensure medium exchange between the halves. After a 4 hour incubation, the bacteria were plated on minimal medium. Davis discovered that when the two auxotrophic strains were separated from each other by the fine filter, gene transfer could not take place. Therefore direct contact was required for the recombination that Lederberg and Tatum had observed.

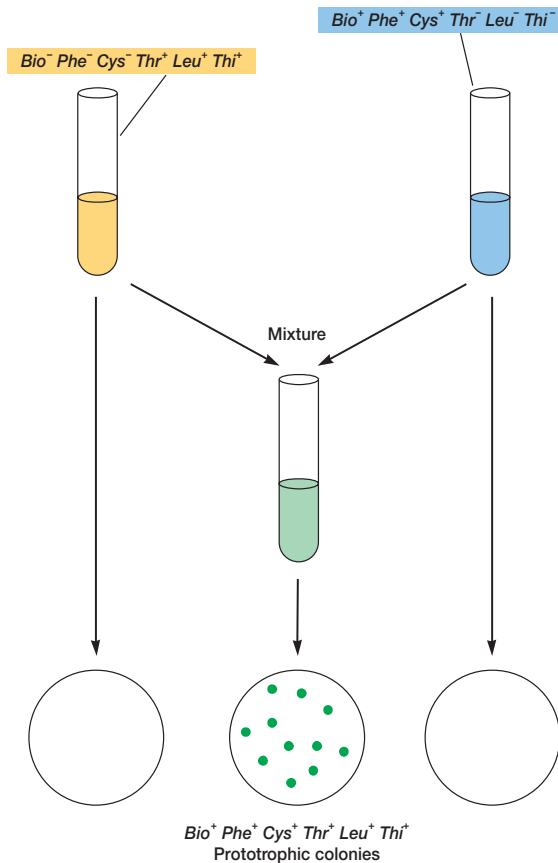


Figure 13.12 Evidence for Bacterial Conjugation. Lederberg and Tatum's demonstration of genetic recombination using triple auxotrophs. See text for details.

$F^+ \times F^-$ Mating

In 1952 William Hayes demonstrated that the gene transfer observed by Lederberg and Tatum was polar. That is, there were definite donor (F^+) and recipient (F^-) strains, and gene transfer was nonreciprocal. He also found that in $F^+ \times F^-$ mating the progeny were only rarely changed with regard to auxotrophy (that is, bacterial genes were not often transferred), but F^- strains frequently became F^+ .

These results are readily explained in terms of the F factor previously described (figure 13.5). The F^+ strain contains an extrachromosomal F factor carrying the genes for pilus formation and plasmid transfer. During $F^+ \times F^-$ mating or conjugation, the F factor replicates by the rolling-circle mechanism, and a copy

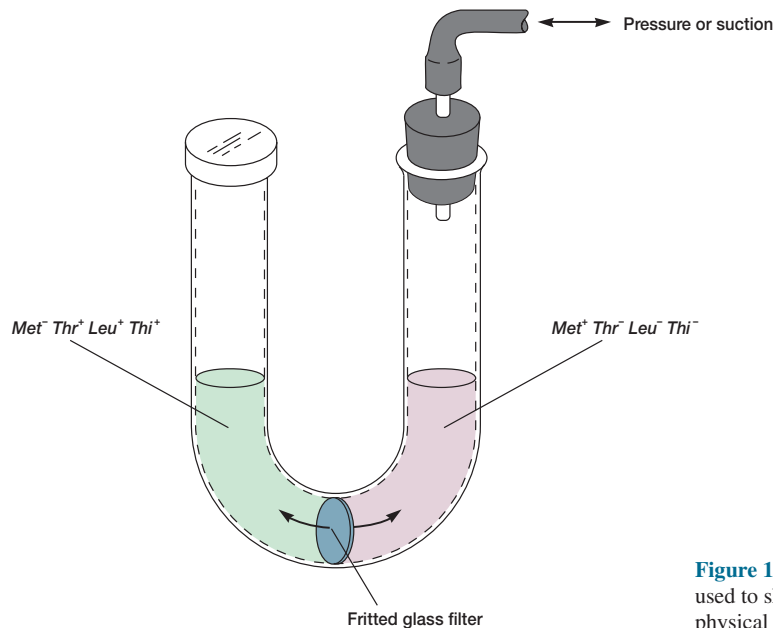


Figure 13.13 The U-Tube Experiment. The U-tube experiment used to show that genetic recombination by conjugation requires direct physical contact between bacteria. See text for details.

moves to the recipient (**figure 13.14a**). The entering strand is copied to produce double-stranded DNA. Because bacterial chromosome genes are rarely transferred with the independent F factor, the recombination frequency is low. It is still not completely clear how the plasmid moves between bacteria. The **sex pilus** or F pilus joins the donor and recipient and may contract to draw them together. The channel for DNA transfer could be either the hollow F pilus or a special conjugation bridge formed upon contact. [The rolling-circle mechanism of DNA replication \(p. 236\)](#)

Although most research on plasmids and conjugation has been done using *E. coli* and other gram-negative bacteria, self-transmissible plasmids are present in gram-positive bacterial genera such as *Bacillus*, *Streptococcus*, *Enterococcus*, *Staphylococcus*, and *Streptomyces*. Much less is known about these systems. It appears that fewer transfer genes are involved, possibly because a sex pilus does not seem to be required for plasmid transfer. For example, *Enterococcus faecalis* recipient cells release short peptide chemical signals that activate transfer genes in donor cells containing the proper plasmid. Donor and recipient cells directly adhere to one another through special plasmid-encoded proteins released by the activated donor cell. Plasmid transfer then occurs.

Hfr Conjugation

Because certain donor strains transfer bacterial genes with great efficiency and do not usually change recipient bacteria to donors, a second type of conjugation must exist. The F factor is an episome and can integrate into the bacterial chromosome at several different locations by recombination between homologous insertion se-

quences present on both the plasmid and host chromosomes. When integrated, the F plasmid's *tra* operon is still functional; the plasmid can direct the synthesis of pili, carry out rolling-circle replication, and transfer genetic material to an F^- recipient cell. Such a donor is called an **Hfr strain** (for high frequency of recombination) because it exhibits a very high efficiency of chromosomal gene transfer in comparison with F^+ cells. DNA transfer begins when the integrated F factor is nicked at its site of transfer origin (**figure 13.14b**). As it is replicated, the chromosome moves through the pilus or conjugation bridge connecting the donor and recipient. Because only part of the F factor is transferred at the start (the initial break is within the F plasmid), the F^- recipient does not become F^+ unless the whole chromosome is transferred. Transfer is standardized at 100 minutes in *E. coli*, and the connection usually breaks before this process is finished. Thus a complete F factor usually is not transferred, and the recipient remains F^- .

As mentioned earlier, when an Hfr strain participates in conjugation, bacterial genes are frequently transferred to the recipient. Gene transfer can be in either a clockwise or counterclockwise direction around the circular chromosome, depending on the orientation of the integrated F factor. After the replicated donor chromosome enters the recipient cell, it may be degraded or incorporated into the F^- genome by recombination.

F' Conjugation

Because the F plasmid is an episome, it can leave the bacterial chromosome. Sometimes during this process the plasmid makes an error in excision and picks up a portion of the chromosomal

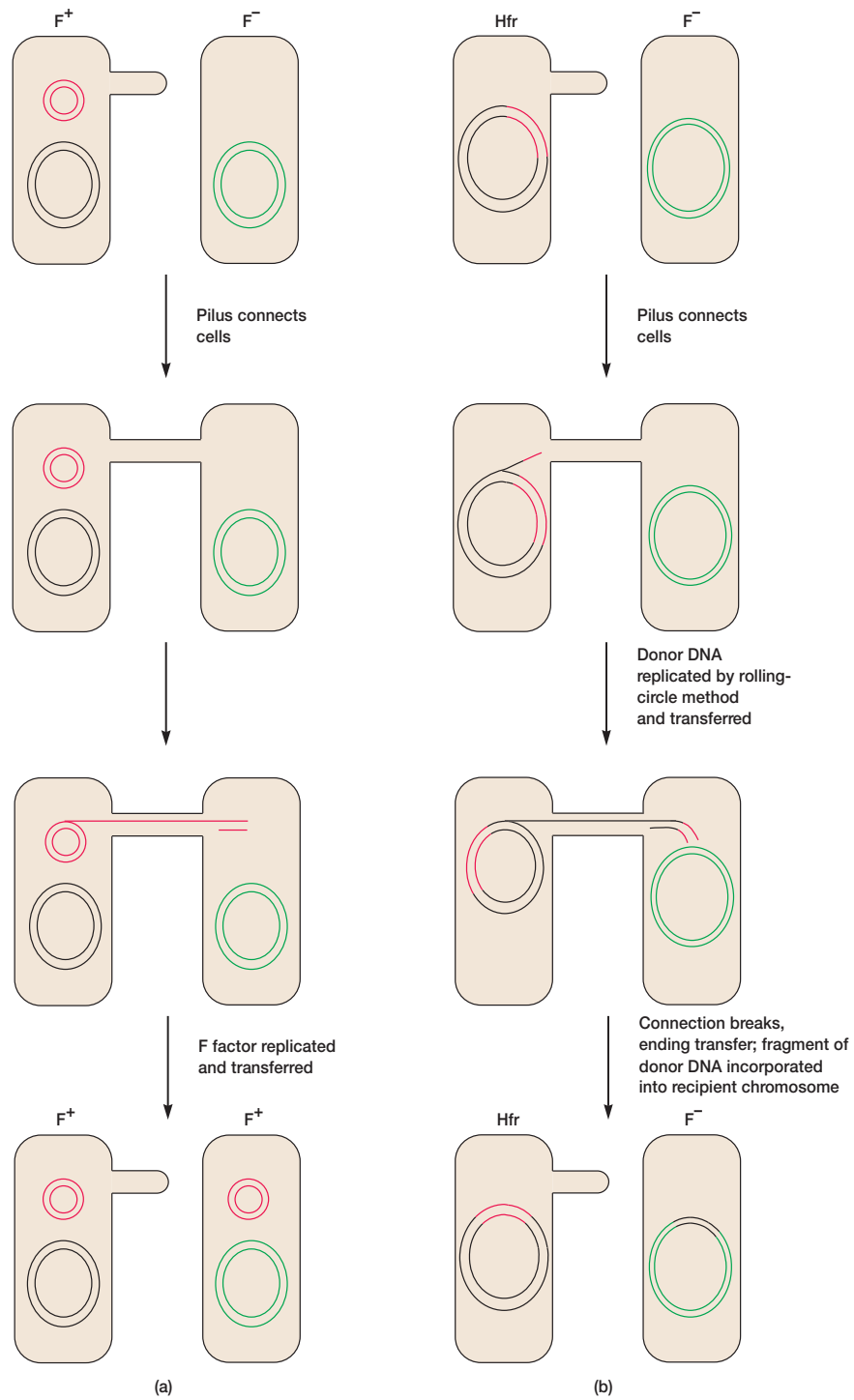


Figure 13.14 The Mechanism of Bacterial Conjugation. (a) $F^+ \times F^-$ mating. (b) $Hfr \times F^-$ mating (the integrated F factor is shown in red).